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Kwon et al.

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(54) **MEMORY DEVICE INCLUDING PROCESSING CIRCUIT, AND ELECTRONIC DEVICE INCLUDING SYSTEM ON CHIP AND MEMORY DEVICE**

(58) **Field of Classification Search**
CPC G06F 13/1668; G11C 8/10; G11C 7/10; H01L 25/0657; H01L 2225/06541
See application file for complete search history.

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(56) **References Cited**

U.S. PATENT DOCUMENTS

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7,999,367 B2 8/2011 Kang et al.
8,031,505 B2 10/2011 Kang et al.
(Continued)

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FOREIGN PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

CA 2497518 A 3/2004
JP 2004-296088 A 10/2004

OTHER PUBLICATIONS

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(Continued)

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Zheng Wei et al., "Overview of Coupler Buffer Device for Guangzhou-Shenzhen-Hong Kong Electric Multiple Unit (EMU)", EMU Technology—Railway Technical Innovation, Chinese Library Classification No. U266.2, Doc Code: B, Article No. 1672-061X(2014)04-0019-08, 2014, DOI:10.19550/j.issn.1672-061x.2014.04.004, pp. 19-26 (47 pages total).

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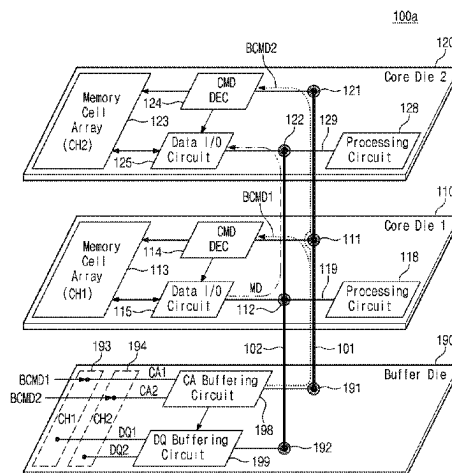
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(57) **ABSTRACT**

A memory device includes a buffer die configured to receive a first broadcast command and a second broadcast command from an external device; and a plurality of core dies stacked on the buffer die. The plurality of core dies include: a first core die including a first processing circuit, a first memory cell array, a first command decoder configured to decode the first broadcast command, and a first data input/output circuit configured to output data of the first memory cell array to a common data input/output bus under control of the first command decoder; and a second core die including a second processing circuit, a second memory cell array, a second command decoder configured to decode the second broadcast command, and a second data input/output circuit con-

(Continued)



figured to receive the data of the first memory cell array through the common data input/output bus under control of the second command decoder.

20 Claims, 17 Drawing Sheets

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continuation of application No. 16/934,497, filed on Jul. 21, 2020, now Pat. No. 11,301,399.

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G11C 8/10 (2006.01)

(56) References Cited

U.S. PATENT DOCUMENTS

8,174,115	B2	5/2012	Chung
8,253,227	B2	8/2012	Osada et al.
8,384,432	B2	2/2013	Kondo
8,599,641	B2	12/2013	Yoko et al.
8,619,486	B2	12/2013	Hayashi et al.
8,693,277	B2	4/2014	Hayashi
8,773,939	B2	7/2014	Oh et al.
9,013,908	B2	4/2015	Chang et al.
9,230,609	B2	1/2016	Frans
9,558,143	B2	1/2017	Leidel
9,851,401	B2	12/2017	Kim et al.
9,971,541	B2	5/2018	Lea et al.
9,997,232	B2	6/2018	Murphy

10,062,428	B2	8/2018	Lee
10,083,722	B2	9/2018	Oh et al.
10,090,038	B2	10/2018	Shin
10,163,469	B2	12/2018	Kondo et al.
10,236,038	B2	3/2019	Lea et al.
10,365,325	B2	7/2019	Dono
10,410,685	B2 *	9/2019	Oh G11C 29/48
10,446,200	B2	10/2019	Hollis
10,459,871	B2	10/2019	Funahashi et al.
10,553,261	B2	2/2020	Kim
10,614,859	B2	4/2020	Vogelsang
10,622,088	B2 *	4/2020	Choi G11C 29/4401
10,748,601	B2	8/2020	Kim et al.
10,790,266	B2	9/2020	Ito et al.
10,811,057	B1	10/2020	Yoshida et al.
10,916,489	B1	2/2021	Nishioka et al.
11,004,477	B2	5/2021	Shibata
11,301,399	B2	4/2022	Kwon et al.
2012/0051113	A1	3/2012	Choi et al.
2013/0162275	A1	6/2013	Kurihara et al.
2016/0357630	A1	12/2016	Kang et al.
2019/0057050	A1	2/2019	Mathuriya et al.
2019/0146788	A1 *	5/2019	Kim G11C 11/54 712/221
2019/0205273	A1	7/2019	Kavalieros et al.
2019/0303042	A1 *	10/2019	Kim H01L 25/18
2020/0210296	A1	7/2020	Kang et al.

OTHER PUBLICATIONS

Seon Wook Kim et al., "Design and Implementation of Display Stream Compression Decoder With Line Buffer Optimization", IEEE Transactions On Consumer Electronics, vol. 65, No. 3, Aug. 2019, pp. 322-328 (7 pages total).

* cited by examiner

FIG. 1

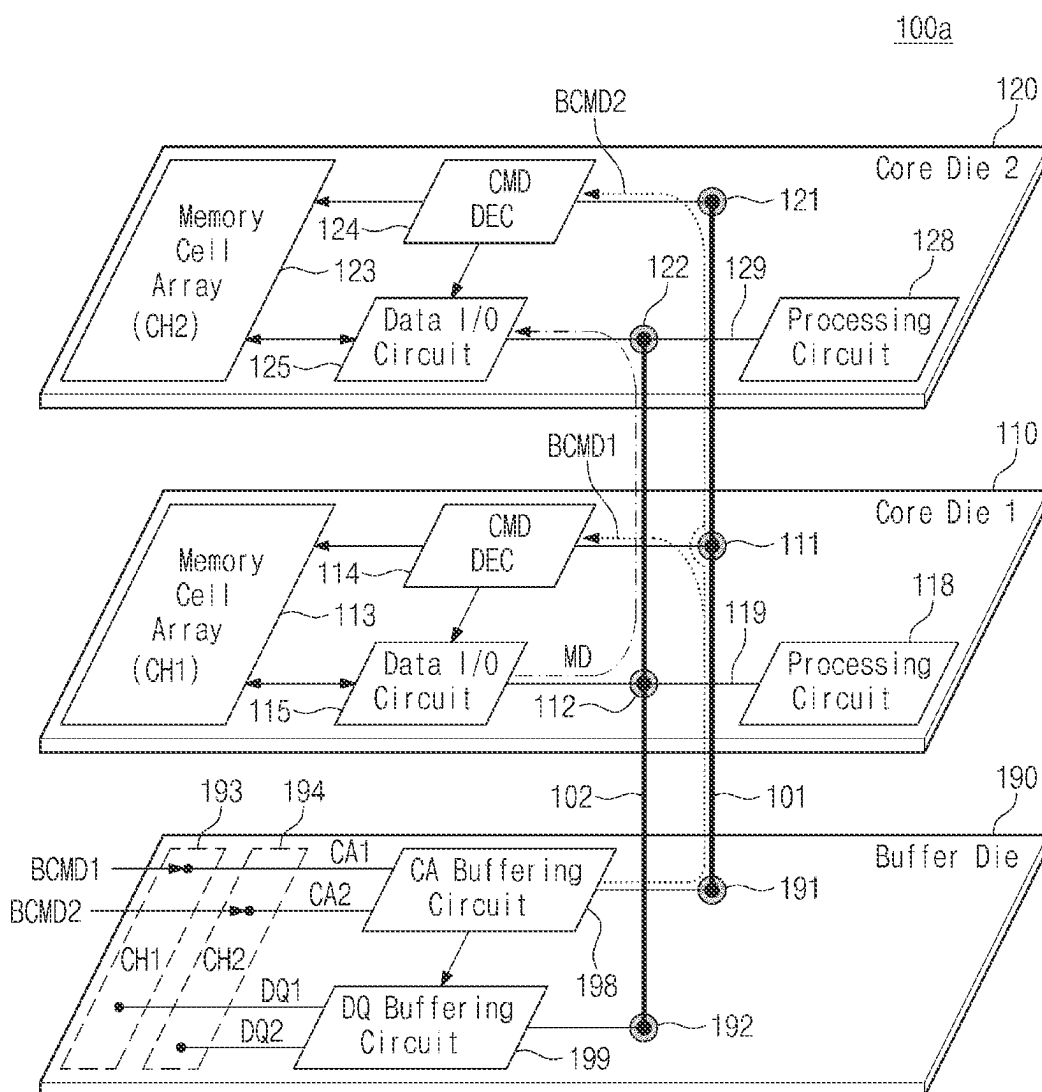


FIG. 2

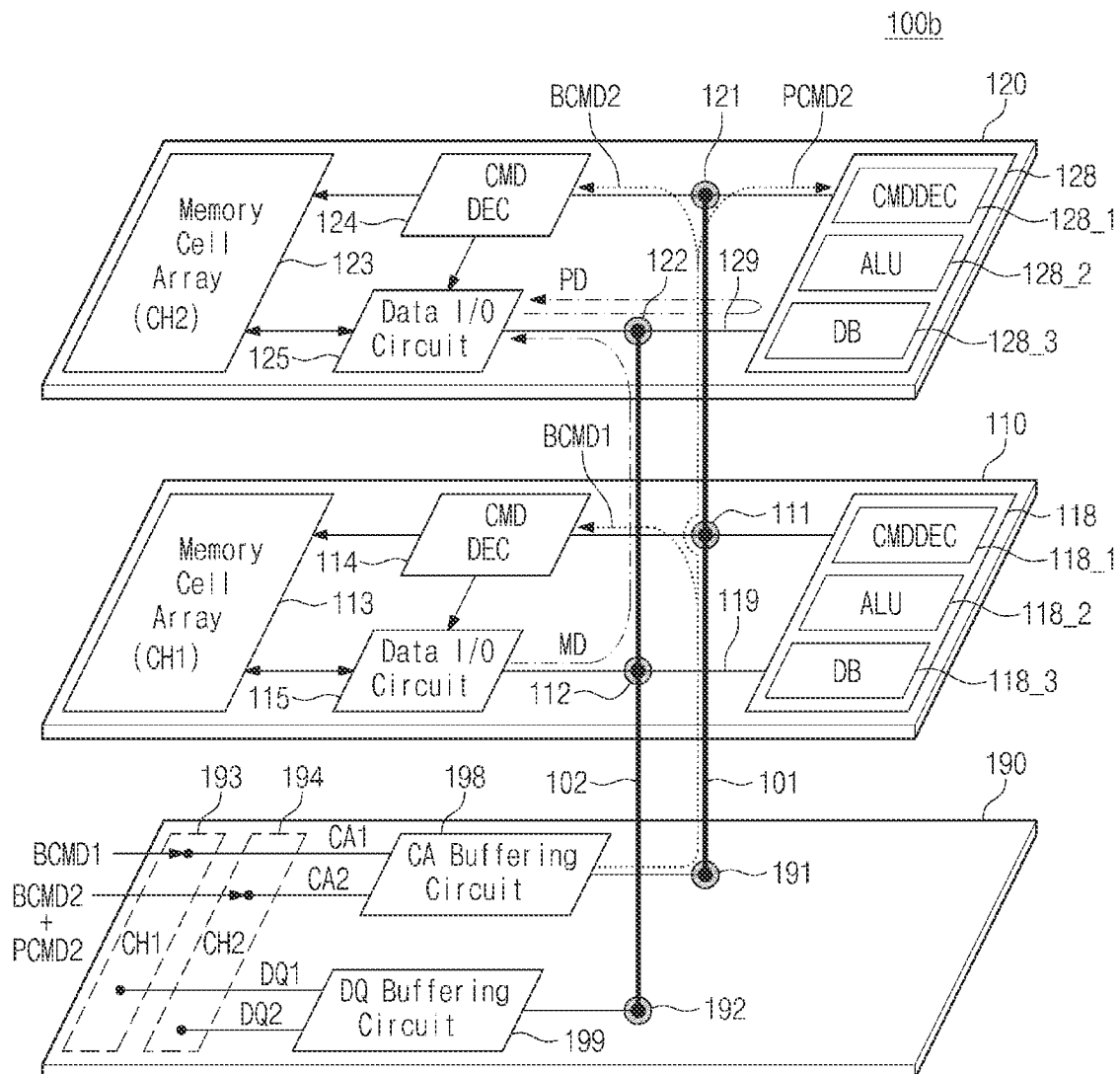


FIG. 3

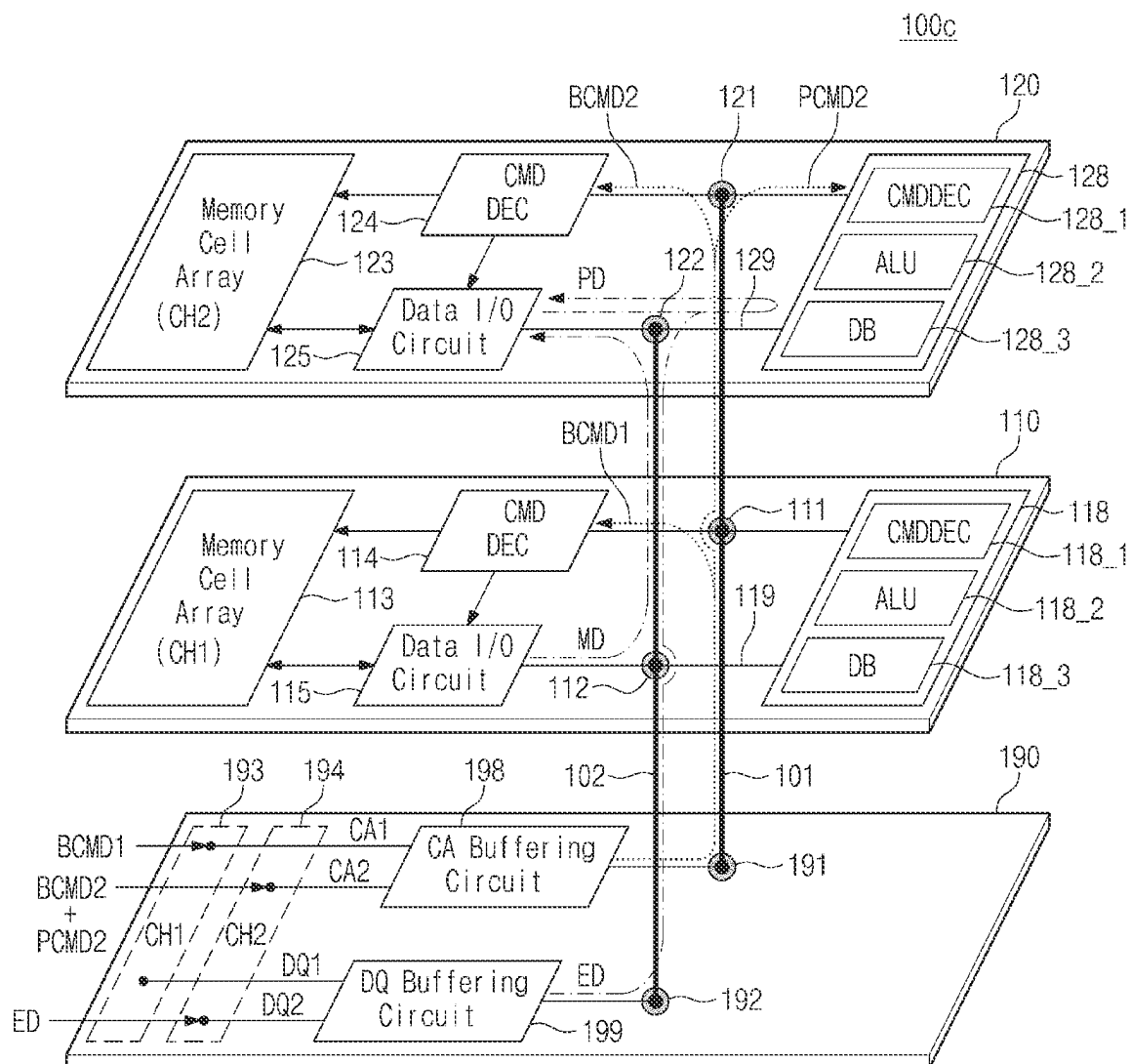


FIG. 4

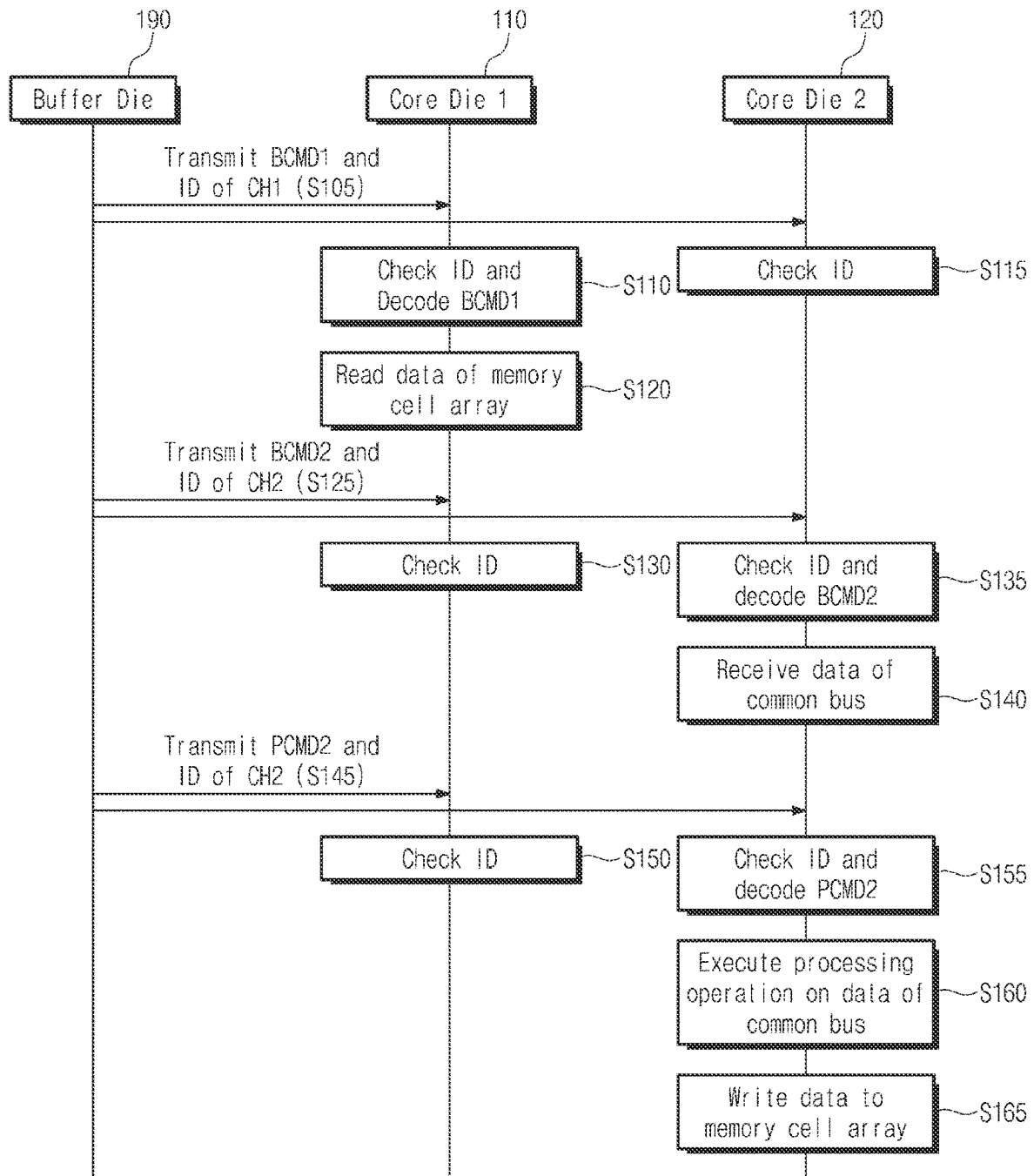


FIG. 5

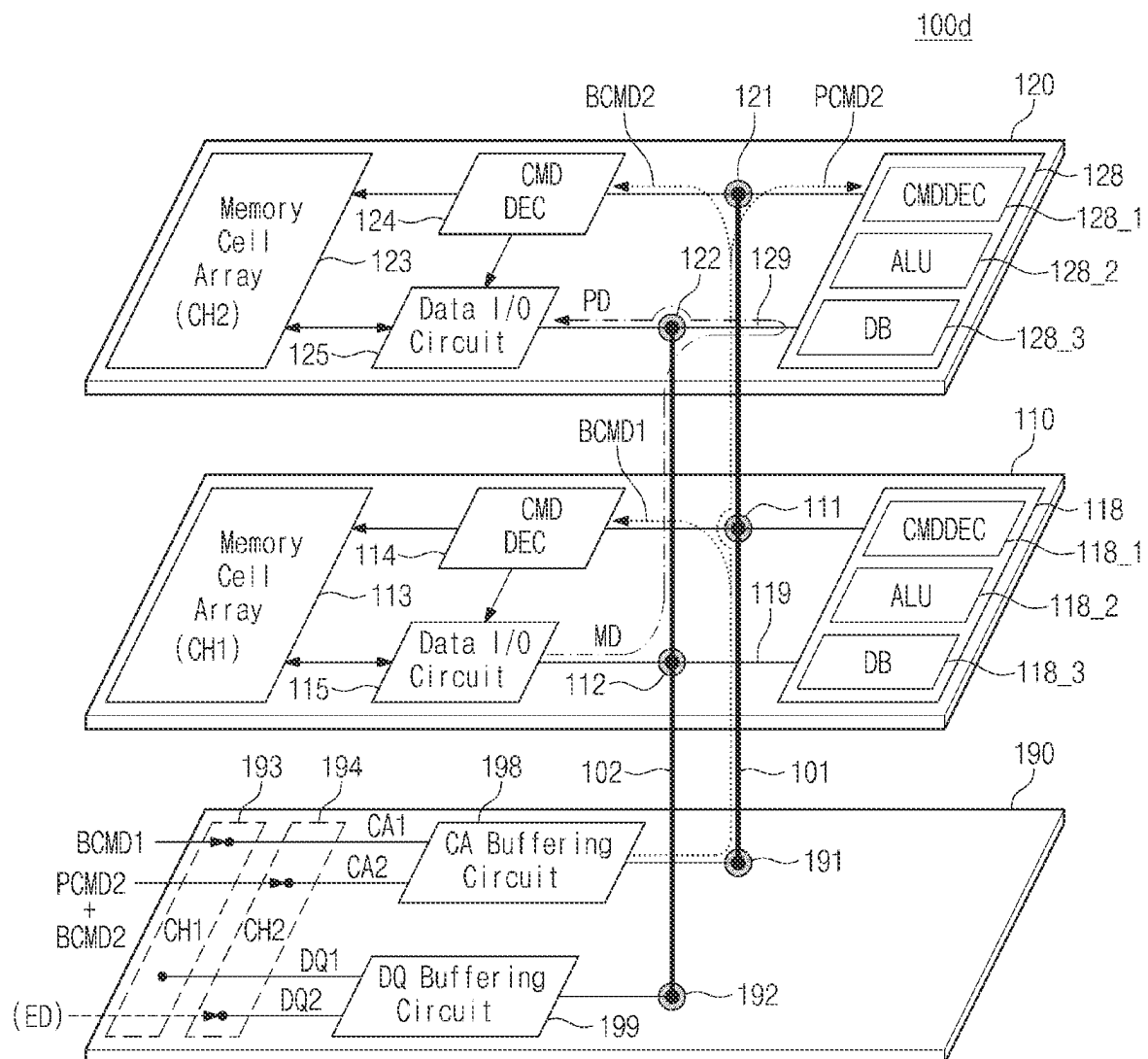


FIG. 6

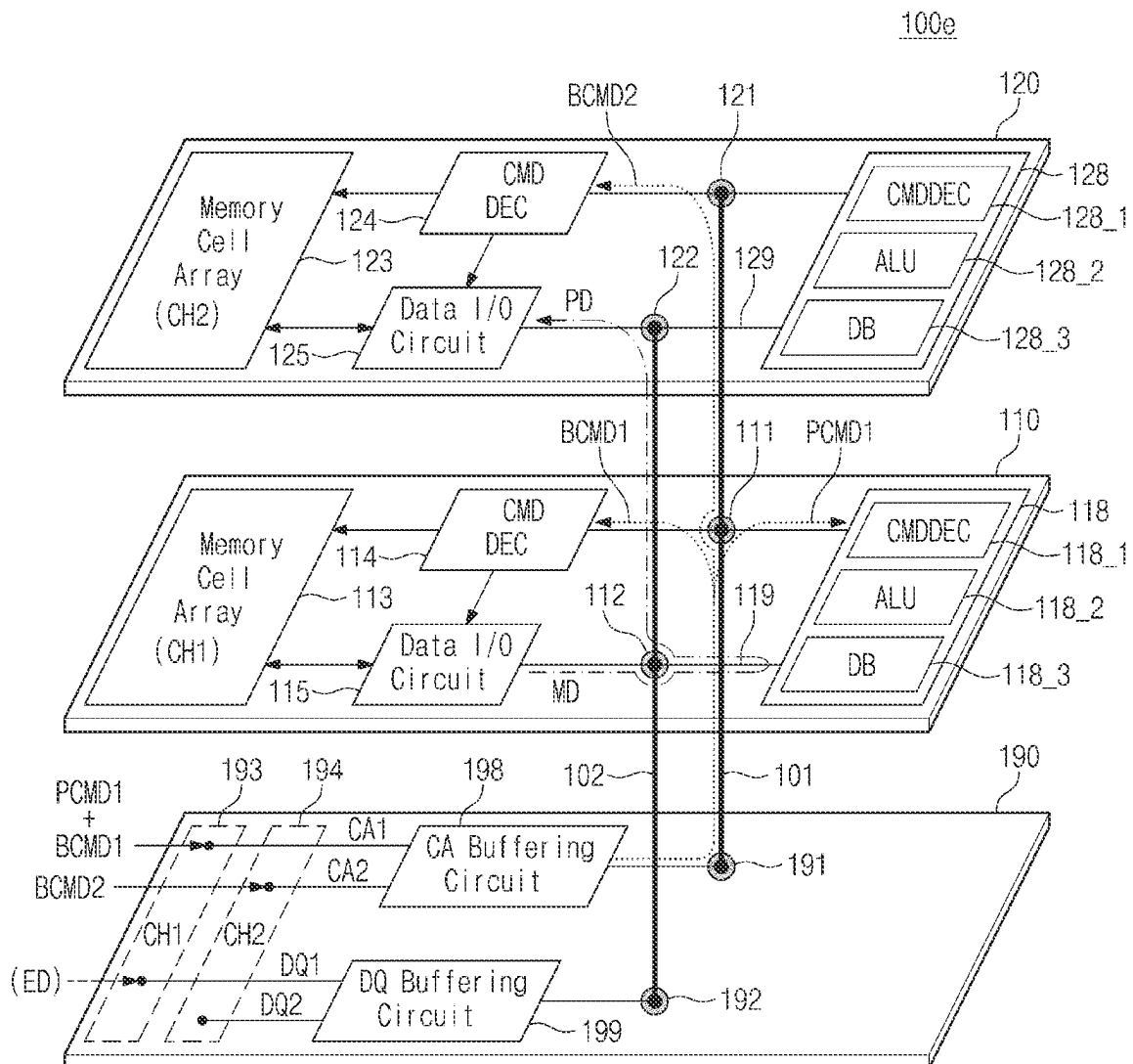


FIG. 7

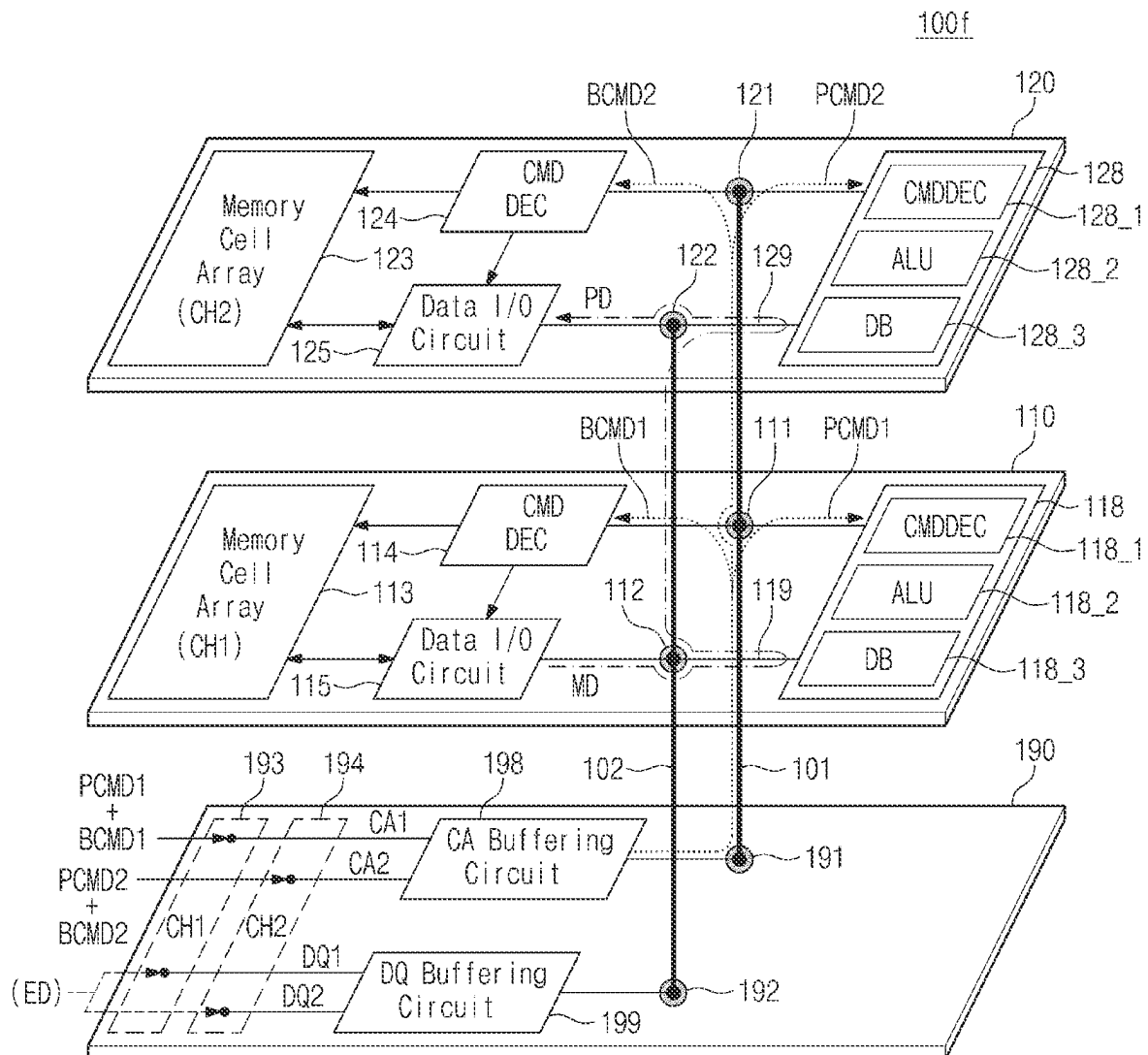


FIG. 8

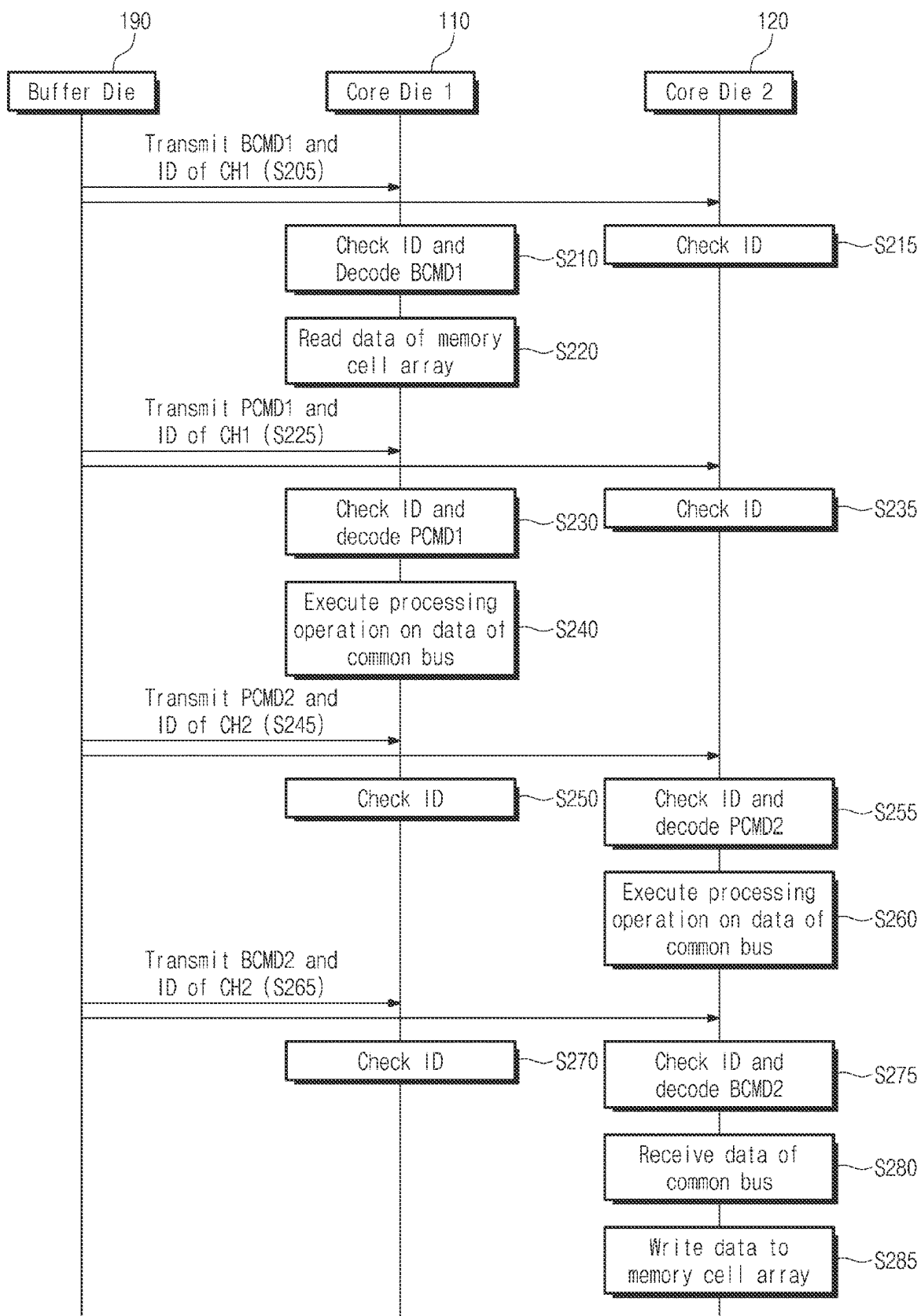


FIG. 9

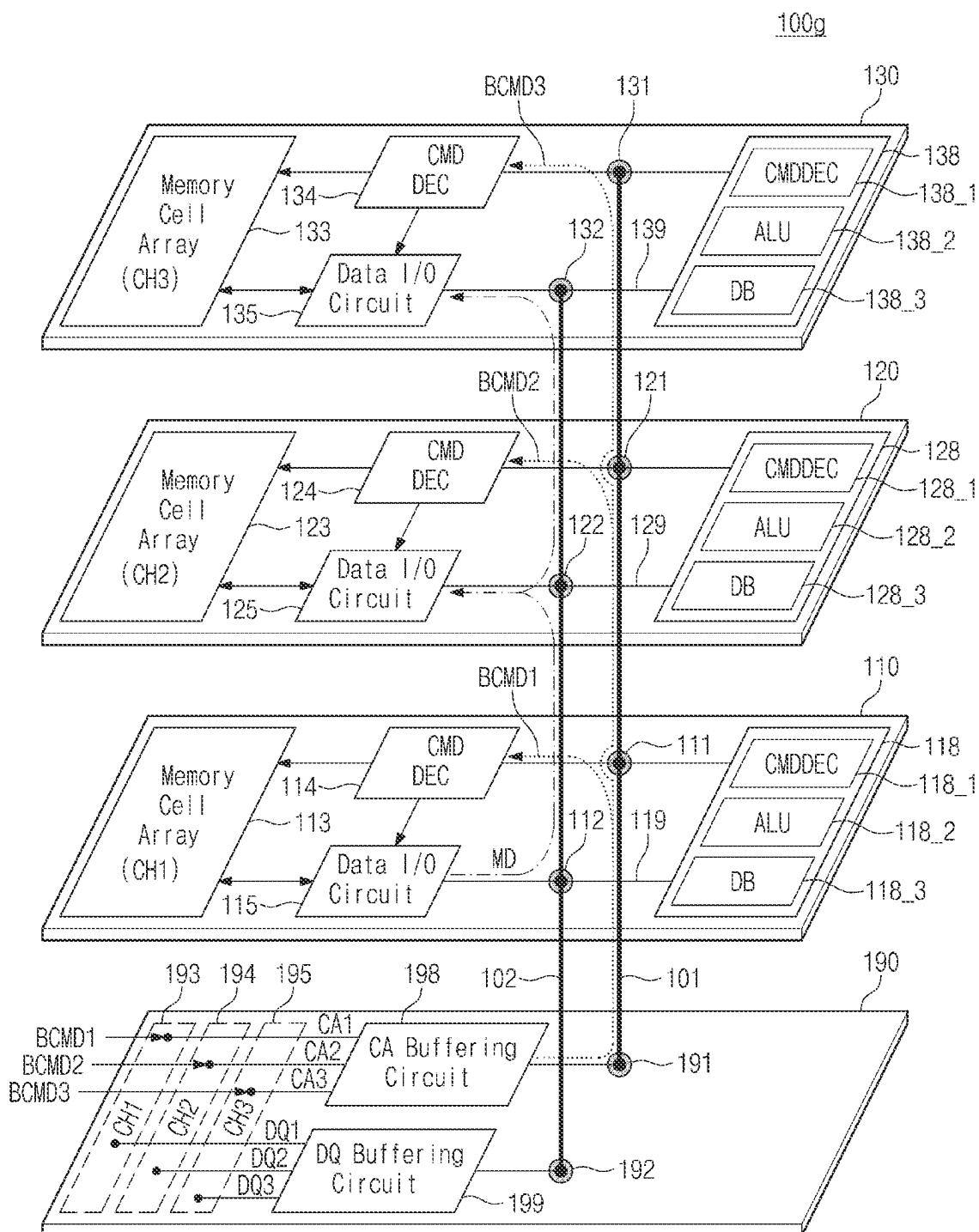


FIG. 10

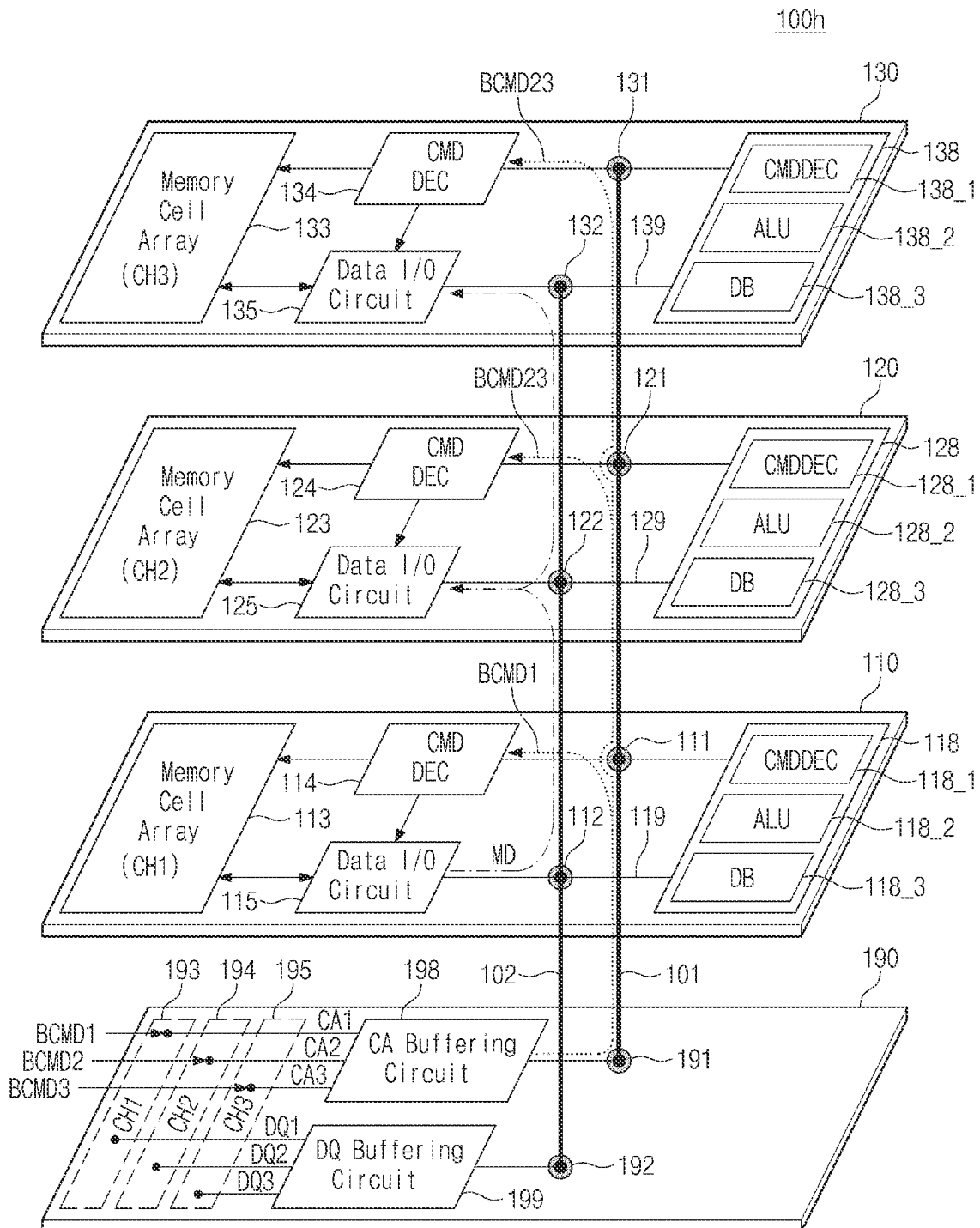


FIG. 11

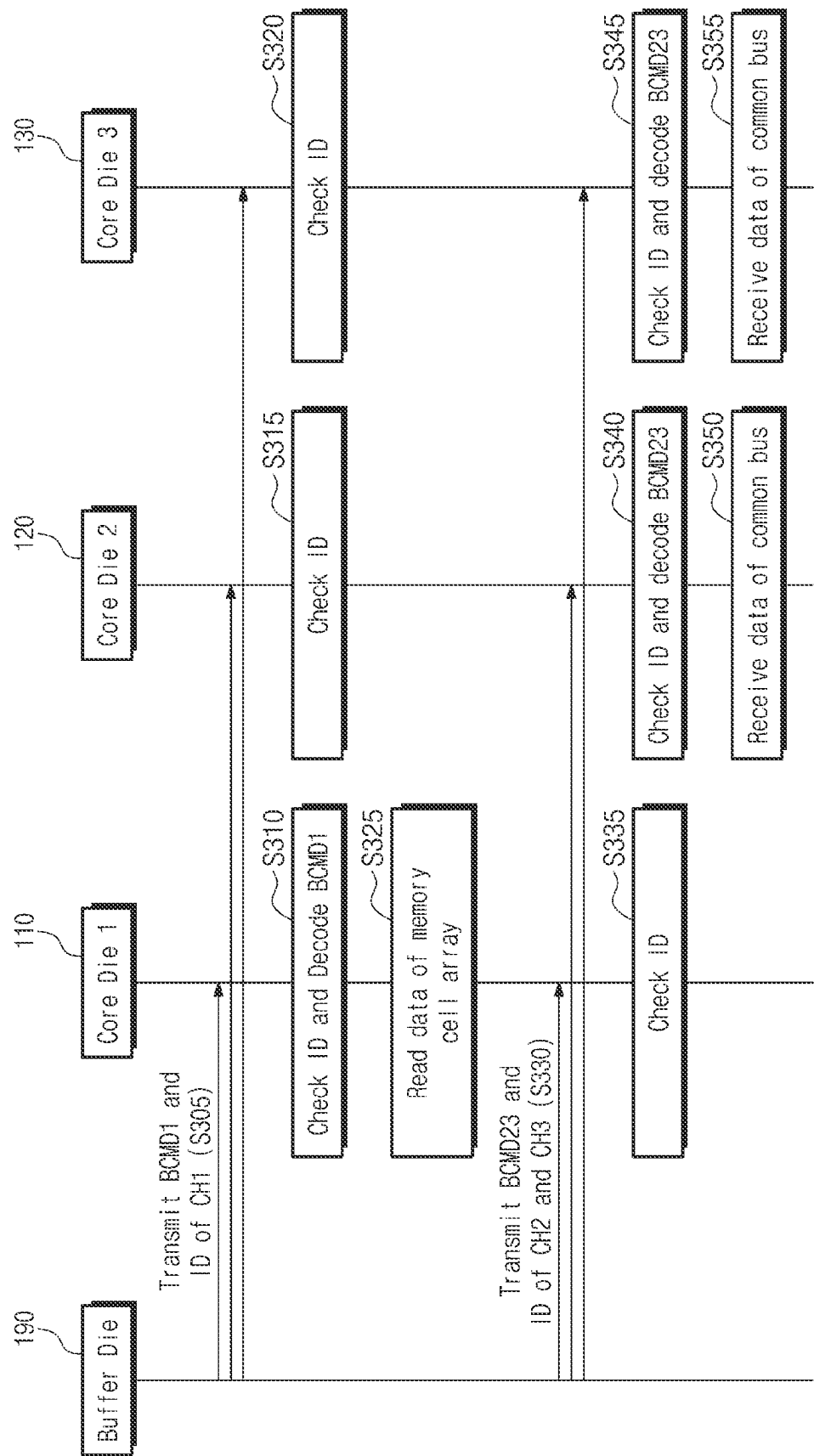


FIG. 12

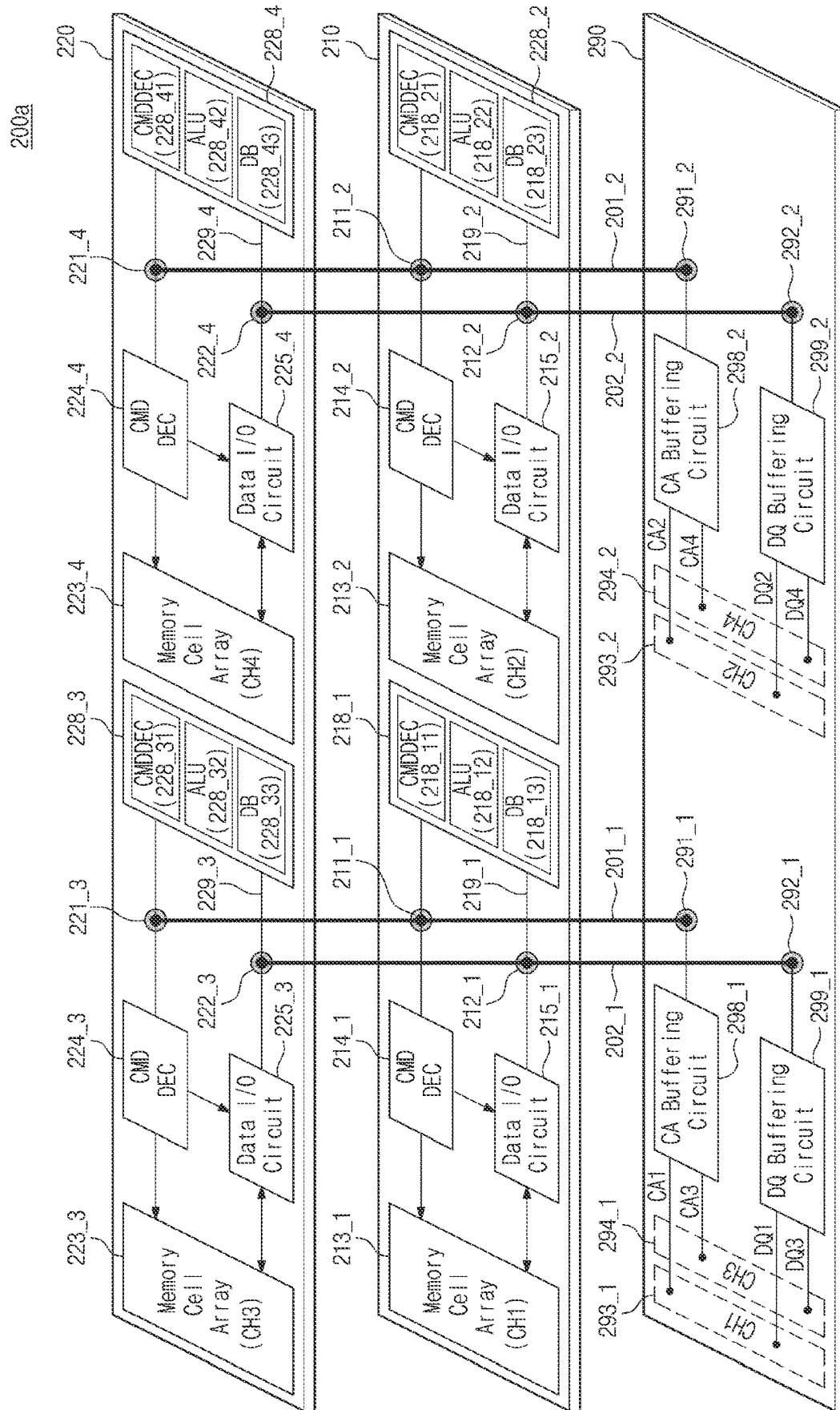


FIG. 13

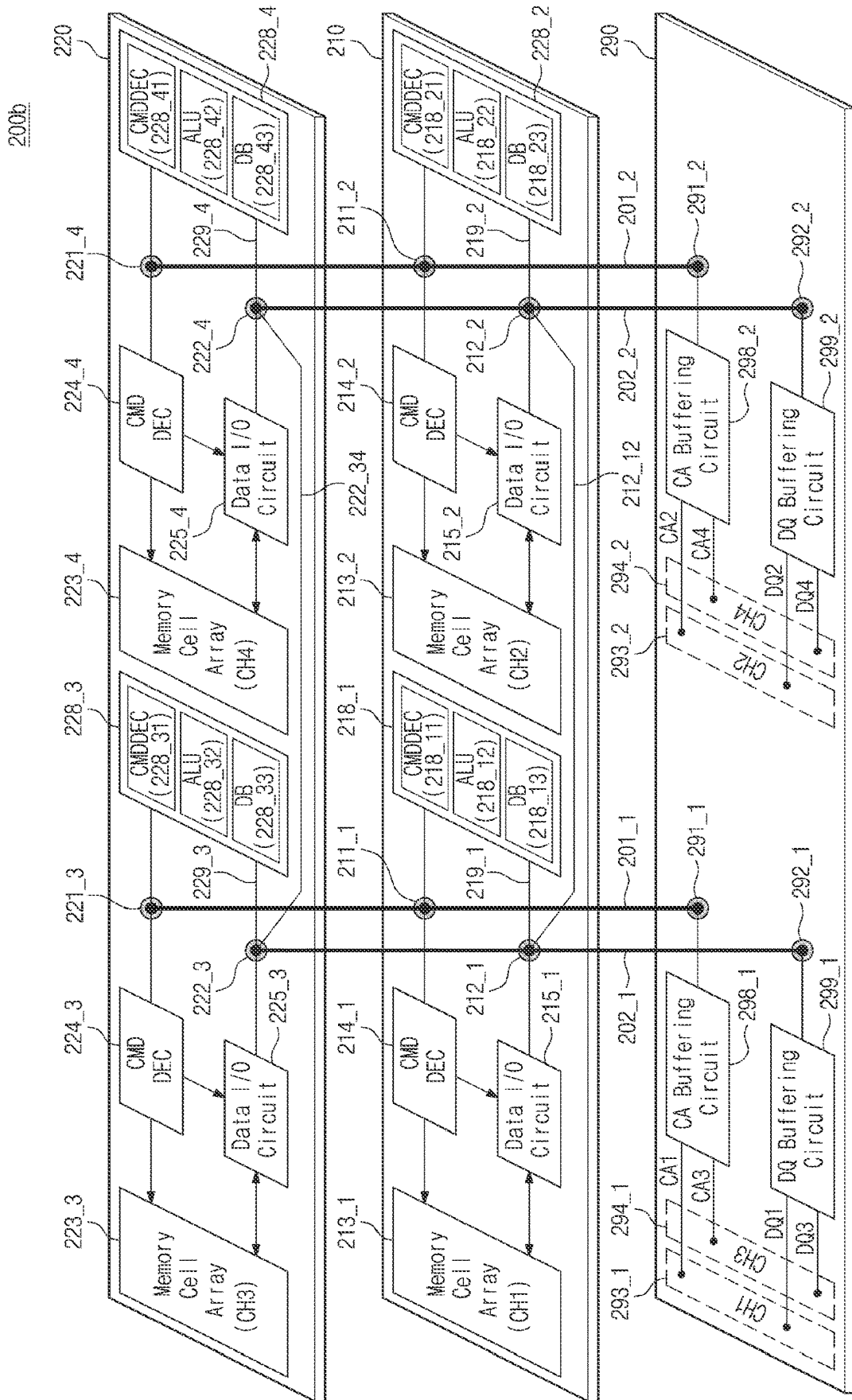


FIG. 14

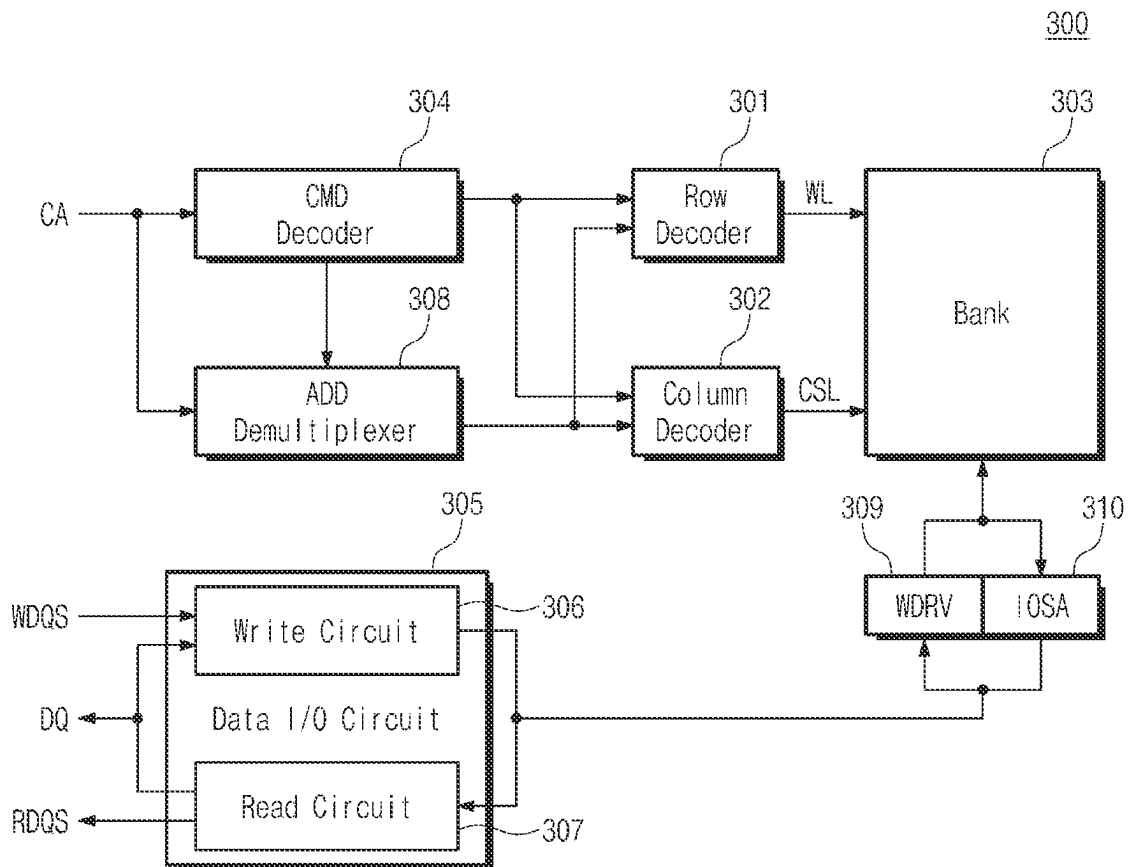


FIG. 15

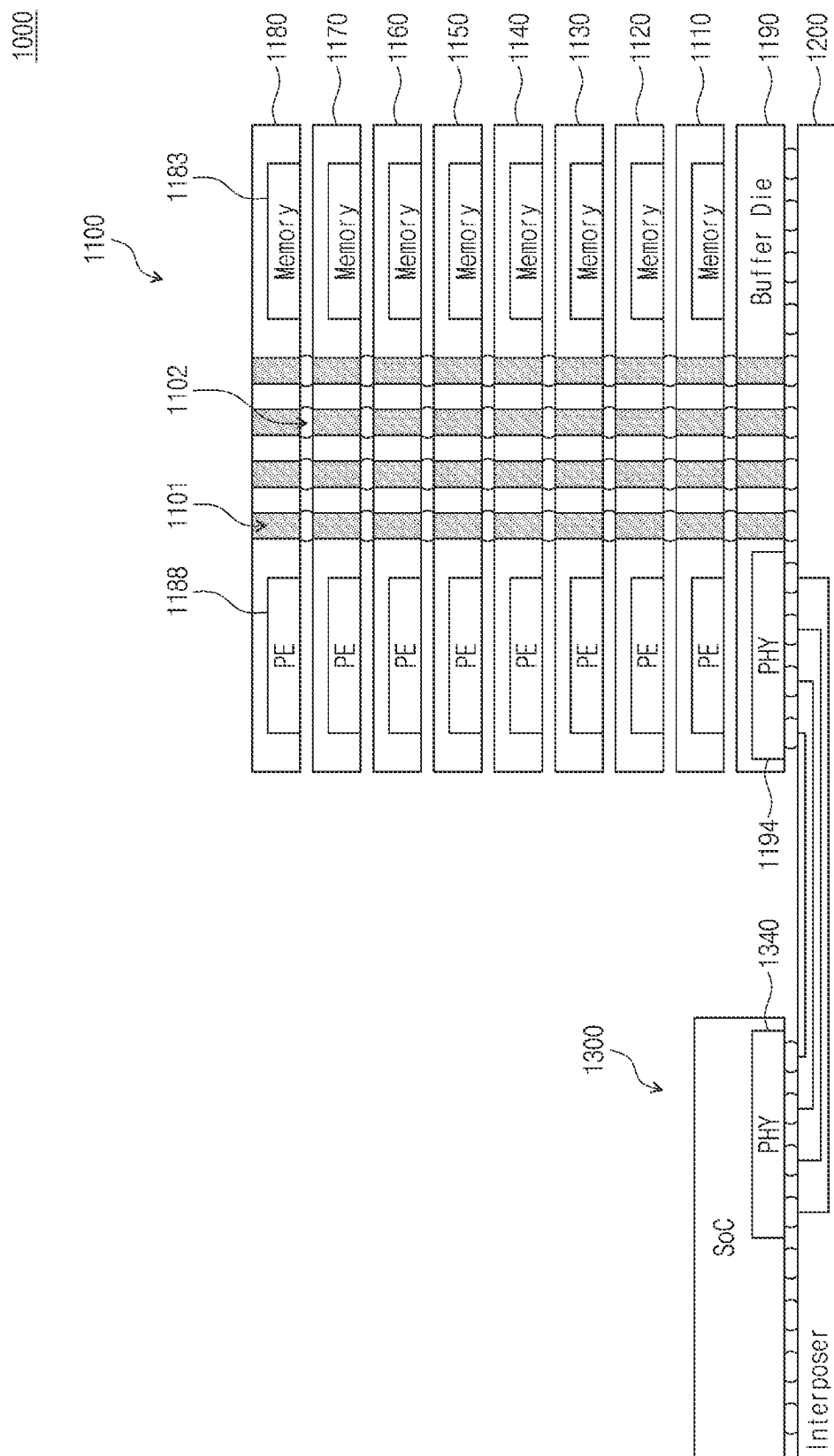


FIG. 16

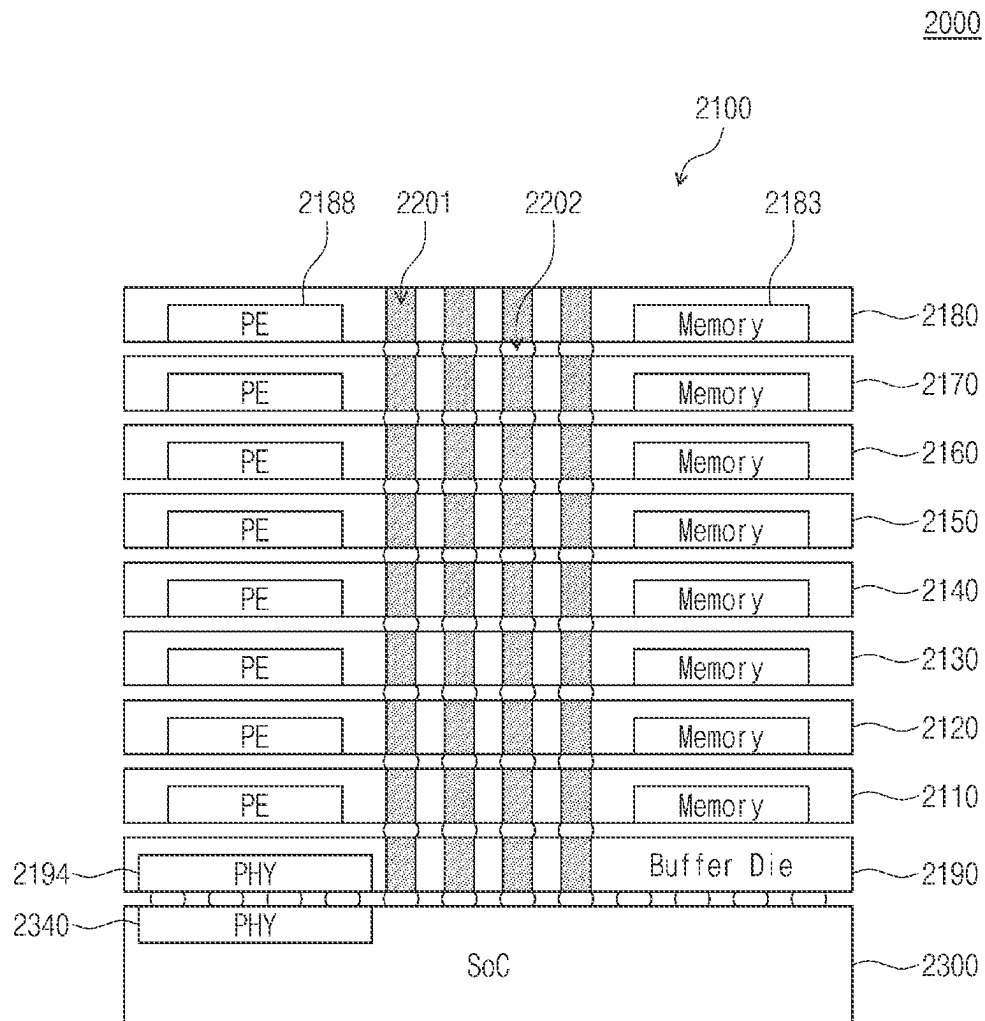
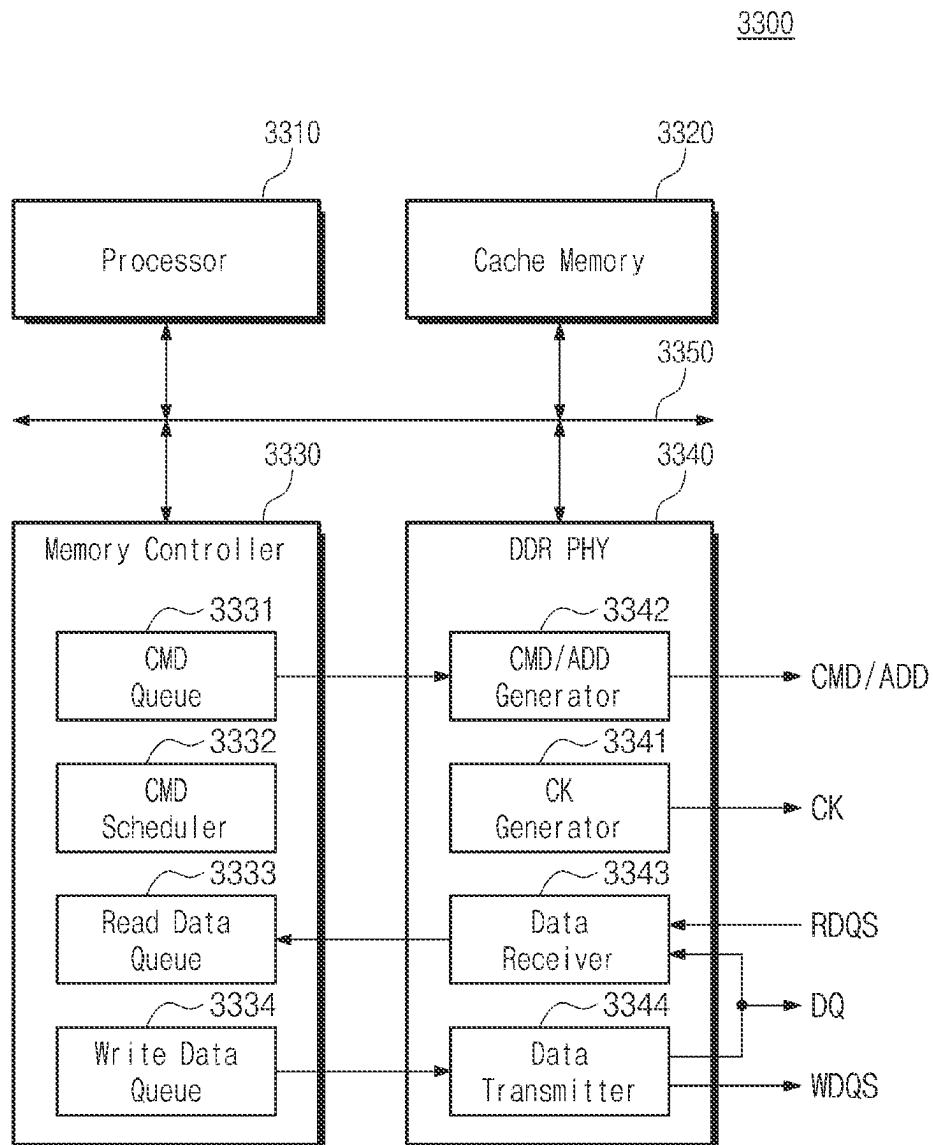


FIG. 17



MEMORY DEVICE INCLUDING PROCESSING CIRCUIT, AND ELECTRONIC DEVICE INCLUDING SYSTEM ON CHIP AND MEMORY DEVICE

CROSS-REFERENCE TO THE RELATED APPLICATION

This application is a continuation application of U.S. patent application Ser. No. 17/685,987 filed Mar. 3, 2022, now U.S. Pat. No. 11,860,803 issued Jan. 2, 2024, which is a continuation application of U.S. patent application Ser. No. 16/934,497, now U.S. Pat. No. 11,301,399 issued Apr. 12, 2022, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2019-0150924 filed on Nov. 22, 2019, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

Methods and apparatuses consistent with example embodiments relate to a memory device including a processing circuit, and more particularly, relate to an electronic device including a system on chip and a memory device.

A plurality of semiconductor dies may be stacked to increase the degree of integration of a memory device. A memory device of a three-dimensional structure may process a large amount of data at a high speed. To implement the three-dimensional structure, a through silicon via (TSV) may be used to stack a plurality of semiconductor dies. As data processing speed increases, the separation of a processor from a memory causes a latency of data that are transmitted between the processor and the memory.

SUMMARY

Example embodiments provide a memory device including a processing circuit, and an electronic device including a system on chip and a memory device.

According to an example embodiment, a memory device includes a buffer die configured to receive a first broadcast command and a second broadcast command from an external device; and a plurality of core dies stacked on the buffer die. The plurality of core dies include: a first core die including a first processing circuit, a first memory cell array, a first command decoder configured to decode the first broadcast command, and a first data input/output circuit configured to output data of the first memory cell array to a common data input/output bus under control of the first command decoder; and a second core die including a second processing circuit, a second memory cell array, a second command decoder configured to decode the second broadcast command, and a second data input/output circuit configured to receive the data of the first memory cell array through the common data input/output bus under control of the second command decoder.

According to an example embodiment, a memory device includes a buffer die configured to receive first to fourth broadcast commands from an external device through first to fourth channels, respectively; a first core die including a first processing circuit and a first memory cell array allocated to the first channel, a second processing circuit and a second memory cell array allocated to the second channel, a first data input/output circuit configured to output data of the first memory cell array to a first common data input/output bus under control of a first command decoder according to the

first broadcast command, and a second data input/output circuit configured to output data of the second memory cell array to a second common data input/output bus under control of a second command decoder according to the second broadcast command; and a second core die including a third processing circuit and a third memory cell array allocated to the third channel, a fourth processing circuit and a fourth memory cell array allocated to the fourth channel, a third data input/output circuit configured to receive the data of the first common data input/output bus under control of a third command decoder according to the third broadcast command, and a fourth data input/output circuit configured to receive the data of the second common data input/output bus under control of a fourth command decoder according to the fourth broadcast command.

According to an example embodiment, an electronic device includes a memory device that includes a buffer die, a first core die including a first processing circuit and a first memory cell array allocated to a first channel, and a second core die including a second processing circuit and a second memory cell array allocated to a second channel; and a system on chip configured to transmit a first broadcast command and a second broadcast command requesting movement of data from the first core die to the second core die to the buffer die of the memory device through the first channel and the second channel, respectively.

BRIEF DESCRIPTION OF THE FIGURES

The above and other aspects, features and advantages will become apparent from the following detailed description of example embodiments, taken in conjunction with the accompanying drawings, in which:

FIGS. 1 to 3 illustrate block diagrams of memory devices according to example embodiments.

FIG. 4 is a flowchart illustrating an operation method of memory devices according to an example embodiment.

FIGS. 5 to 7 illustrate block diagrams of memory devices according to example embodiments.

FIG. 8 is a flowchart illustrating an operation method of a memory device according to an example embodiment.

FIGS. 9 and 10 illustrate block diagrams of memory devices according to example embodiments.

FIG. 11 is a flowchart illustrating an operation method of a memory device according to an example embodiment.

FIGS. 12 and 13 illustrate block diagrams of memory devices according to example embodiments.

FIG. 14 illustrates a block diagram of a memory area of a core die included in a memory device according to an example embodiment.

FIGS. 15 and 16 illustrate electronic devices according to example embodiments.

FIG. 17 illustrates a block diagram of system on chips according to an example embodiment.

DETAILED DESCRIPTION

FIG. 1 illustrates a block diagram of a memory device according to an example embodiment. A memory device 100a may relate to a processor in memory (PIM) or a function in memory (FIM) and may further execute a data processing operation in addition to reading and writing data. The memory device 100a may correspond to a computational memory device including a random access memory (RAM) and a processing element (PE) integrated in the same die.

The memory device **100a** may include core dies **110** and **120** and a buffer die **190**. For example, each of the core dies **110** and **120** may be also referred to a “memory die”, a “PIM die”, an “FIM die”, or a “slave die”, and the buffer die **190** may be also referred to as an “interface die”, a “logic die”, or a “master die”. A die may be also referred to as a “chip”. The core die **110** may be stacked on the buffer die **190**, and the core die **120** may be stacked on the core die **110**. The memory device **100a** may have a three-dimensional memory structure in which the plurality of dies **110**, **120**, and **190** are stacked.

The memory device **100a** may include a common command and address bus **101** including paths through which command and address (or command/address) signals **CA1** and **CA2** are transmitted. The common command and address bus **101** may include through silicon vias (TSVs) **111** passing through the core die **110**, TSVs **121** passing through the core die **120**, TSVs **191** passing through the buffer die **190**, and micro bumps (refer to **1102** of FIG. **15**) providing an electrical connection or contact between the TSVs **111**, **121**, and **191**. The memory device **100a** may include a common data input/output bus **102** including paths through which data input/output signals **DQ1** and **DQ2** are transmitted. The common data input/output bus **102** may include TSVs **112** passing through the core die **110**, TSVs **122** passing through the core die **120**, TSVs **192** passing through the buffer die **190**, and micro bumps (refer to **1102** of FIG. **15**) providing an electrical connection or contact between the TSVs **112**, **122**, and **192**. The core dies **110** and **120** and the buffer die **190** may be electrically interconnected through the common command and address bus **101** and the common data input/output bus **102**.

The core die **110** may include the TSVs **111** and **112**, a memory cell array **113** allocated to a channel **CH1**, a command decoder **114**, a data input/output circuit **115**, a processing circuit **118**, and an internal data input/output bus **119**. The channel **CH1** may include access (or communication) paths capable of accessing the memory device **100a**. For example, an external device (e.g., a memory controller, a system on chip (SoC), an application processor (AP), or a host) may access the memory cell array **113** through the channel **CH1**, may transmit commands and addresses associated with the memory cell array **113**, and may exchange data on the memory cell array **113**.

The memory cell array **113** may include memory cells connected to word lines (or rows) and bit lines (or columns). For example, a memory cell may be a dynamic random access (DRAM) cell, a static random access memory (SRAM) cell, a NAND flash memory cell, a NOR flash memory cell, a resistive random access memory (RRAM) cell, a ferroelectric random access memory (FRAM) cell, a phase change random access memory (PRAM) cell, a thyristor random access memory (TRAM) cell, a magnetic random access memory (MRAM) cell, etc.

The command decoder **114** may decode commands transmitted through the buffer die **190** and the common command and address bus **101**. For example, the commands may include the following command associated with the memory cell array **113**: an active command, a precharge command, a write command, and a refresh command; alternatively, the commands may include the following commands associated with an operation mode of the core die **110**: a mode register set command, a mode register write command, and a mode register read command. Additionally, the commands may include a broadcast command requesting a data movement within the core dies **110** and **120** and a processing command requesting a processing operation on data of the core dies

110 and **120**. The broadcast command may be also referred to as a “move command” or a “transfer command”. The command decoder **114** may decode a command and may control the memory cell array **113** and the data input/output circuit **115**.

The data input/output circuit **115** may receive write data through the common data input/output bus **102** and may transmit the write data to the memory cell array **113**. The write data may be written, stored, or programmed to the memory cell array **113**. The write data may be transmitted from the external device through the channel **CH1**, the buffer die **190**, and the common data input/output bus **102**, may be transmitted from the processing circuit **118** through the internal data input/output bus **119**, or may be transmitted from the core die **120** through the common data input/output bus **102**. The data input/output circuit **115** may read data stored in the memory cell array **113** or may output (or transmit) the read data to the common data input/output bus **102**. The read data of the common data input/output bus **102** may be transmitted to the external device through the buffer die **190** and the channel **CH1**, may be transmitted to the core die **120** through the common data input/output bus **102**, or may be transmitted to the processing circuit **118** through the internal data input/output bus **119**.

The processing circuit **118** may perform a processing operation on data of the internal data input/output bus **119** or data of the common data input/output bus **102**. For example, the data of the internal data input/output bus **119** may mean data output from the data input/output circuit **115** to the internal data input/output bus **119**. For example, the data of the common data input/output bus **102** may mean data output from the data input/output circuit **115**, the core die **120**, or the buffer die **190** to the common data input/output bus **102**. The processing circuit **118** may be disposed on the same core die **110** together with the memory cell array **113** and may be referred to as a “PE” or a “processor”. For example, the processing circuit **118** may be accessed through the channel **CH1** capable of accessing the memory cell array **113** and may be allocated to the channel **CH1**.

The paths of the internal data input/output bus **119** may be electrically connected to the paths of the common data input/output bus **102**, respectively. Accordingly, the paths of the internal data input/output bus **119** may be included in the paths of the common data input/output bus **102**. For another example, the paths of the internal data input/output bus **119** may be electrically separated from the paths of the common data input/output bus **102**, respectively. In any case, the data input/output circuit **115** may output the same data to the internal data input/output bus **119** or the common data input/output bus **102**.

The core die **120** may include the TSVs **121** and **122**, a memory cell array **123**, a command decoder **124**, a data input/output circuit **125**, a processing circuit **128**, and an internal data input/output bus **129**. For example, the core die **120** may be stacked on the core die **110**. As in the channel **CH1**, a channel **CH2** may include access (or communication) paths capable of accessing the memory device **100a** and may be independent of the channel **CH1**. For example, the external device may access the memory cell array **123** or the processing circuit **128** through the channel **CH2**. The core die **120** may be implemented to be substantially identical to the core die **110**. Operations of the components **123** to **129** of the core die **120** may be similar or substantially identical to the operations of the components **113** to **119** of the core die **110** except that the core die **120** is accessible through the channel **CH2** and the core die **110** is accessible through the channel **CH1**.

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The buffer die **190** may include the TSVs **191** and **192**, pins **193** and **194**, a command and address buffering circuit **198** and a data input/output buffering circuit **199**. The pins **193** may be allocated to the channel CH1. The pins **193** may include command and address pins receiving command and address signals CA1 transmitted from the external device through the channel CH1 and data input/output pins receiving the data input/output signals DQ1 transmitted from the external device through the channel CH1. Because the data input/output signals DQ1 are bidirectional, the data input/output signals DQ1 may be transmitted from the above data input/output pins to the external device through the channel CH1. Here, "DQ" may mean a data input/output. The pins **194** may be implemented to be substantially identical to the pins **193** except that the pins **194** are allocated to the channel CH2.

The command and address buffering circuit **198** may buffer (i.e., receive and amplify) the command and address signals CA1 transmitted through the pins **193** and may output the command and address signals CA1 to the common command and address bus **101**. The command and address signals CA1 may relate to the core die **110** and may in detail include commands for the memory cell array **113**, addresses (e.g., including a bank address, a row address, and a column address) indicating memory cells of the memory cell array **113**, or commands for the processing circuit **118**. For example, the command and address buffering circuit **198** may output an identification (ID) (or an address) indicating the channel CH1 (or the core die **110**) to the common command and address bus **101** together with the command and address signals CA1. The command decoder **114** may identify the ID of the channel CH1, may decode the command and address signals CA1, and may not decode the command and address signals CA2. As in the above manner, the command and address buffering circuit **198** may output the command and address signals CA2 to the common command and address bus **101**. As in the command decoder **114**, the command decoder **124** may identify the ID of the channel CH2 and may decode the command and address signals CA2.

The data input/output buffering circuit **199** may receive or output the data input/output signals DQ1 through the channel CH1 and the pins **193**. The data input/output buffering circuit **199** may output the data input/output signals DQ1 and may further output the ID of the channel CH1 to the common data input/output bus **102**. The data input/output circuit **115** may identify the ID of the channel CH1 and may receive the data input/output signals DQ1. The data input/output circuit **115** may also receive the data input/output signals DQ1 without identifying the ID of the channel CH1 under control of the command decoder **114**. The data input/output circuit **115** may output the data input/output signals DQ1 to the common data input/output bus **102** and may further output the ID of the channel CH1 to the common data input/output bus **102**. The data input/output buffering circuit **199** may receive the data input/output signals DQ1 through the common data input/output bus **102**. For example, the data input/output buffering circuit **199** may identify the ID of the channel CH1 and may output the data input/output signals DQ1 to the external device through the pins **193** and the channel CH1. For another example, under control of the command and address buffering circuit **198**, the data input/output buffering circuit **199** may output the data input/output signals DQ1 to the external device through the pins **193** and the channel CH1 without identifying the ID. As in the above manner, the data input/output buffering circuit **199** may output the data input/output signals DQ2. As in the data

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input/output circuit **115**, the data input/output circuit **125** may identify the ID of the channel CH2 and output the data input/output signals DQ2.

Referring to FIG. 1, the memory device **100a** may receive broadcast commands BCMD1 and BCMD2 from the external device through the channels CH1 and CH2, respectively. The broadcast commands BCMD1 and BCMD2 may request the movement of data from the memory cell array **113** allocated to the channel CH1 to the memory cell array **123** allocated to the channel CH2. For example, the broadcast command BCMD1 may indicate a start and a start location of the data movement (i.e., the memory cell array **113** allocated to the channel CH1), and the broadcast command BCMD2 may indicate an end (or target) and an end location of the data movement (i.e., the memory cell array **123** allocated to the channel CH2).

The command and address buffering circuit **198** may receive the broadcast command BCMD1 through the channel CH1 and the pins **193**, may generate the ID of the channel CH1, and may output the broadcast command BCMD1 and the ID of the channel CH1 to the common command and address bus **101**. The command decoder **114** may identify the ID of the channel CH1, may decode the broadcast command BCMD1, and may control the memory cell array **113** and the data input/output circuit **115**. The data input/output circuit **115** may receive data stored in memory cells of the memory cell array **113**, which addresses of the broadcast command BCMD1 indicate, and may output data MD to the common data input/output bus **102**. For example, the operations of the memory cell array **113**, the command decoder **114**, and the data input/output circuit **115**, which are performed in response to the broadcast command BCMD1, may correspond to a read operation of the core die **110**, but the data MD may not be output to the external device unlike a normal read operation.

The command and address buffering circuit **198** may receive the broadcast command BCMD2 through the channel CH2 and the pins **194**, may generate the ID of the channel CH2, and may output the broadcast command BCMD2 and the ID of the channel CH2 to the common command and address bus **101**. The command decoder **124** may identify the ID of the channel CH2, may decode the broadcast command BCMD2, and may control the memory cell array **123** and the data input/output circuit **125**. The data input/output circuit **125** may receive the data MD of the common data input/output bus **102**. The received data MD of the data input/output circuit **125** may be transmitted to memory cells of the memory cell array **123**, which addresses of the broadcast command BCMD2 or a next command indicate. For example, the operations of the memory cell array **123**, the command decoder **124**, and the data input/output circuit **125**, which are performed in response to the broadcast command BCMD2, may correspond to a write operation of the core die **120**, but the write data MD may be provided from the core die **110** and not from the external device unlike a normal write operation.

For example, the broadcast command BCMD1 may be transmitted to the memory device **100a** prior to the broadcast command BCMD2. The order of transmitting the broadcast commands BCMD1 and BCMD2 may depend on a start and an end of the data movement. The broadcast command BCMD2 may be transmitted to the memory device **100a** prior to the broadcast command BCMD1. For another example, the broadcast commands BCMD1 and BCMD2 may be simultaneously transmitted to the memory device **100a**. In this case, a bit (or logical value) of a specific signal of the broadcast command BCMD1 may indicate the start of

the data movement, a bit (or logical value) of a specific signal of the broadcast command BCMD2 may indicate the end of the data movement, and the values of both bits may be different from each other.

The memory device 100a according to an example embodiment may support the data movement between the core dies 110 and 120 based on (or by using) the broadcast commands BCMD1 and BCMD2 issued or output from the external device. The external device may set the start and end of the data movement by adjusting the order of transmitting (or issuing or generating) the broadcast commands BCMD1 and BCMD2 or setting the values of the specific bits of the broadcast commands BCMD1 and BCMD2. Because the memory device 100a supports the data movement through the broadcast commands BCMD1 and BCMD2, the external device may execute (or accomplish) the data movement through the broadcast commands BCMD1 and BCMD2 without needing to read data from the core die 110 of the memory device 100a and transmit the read data to the core die 120 of the memory device 100a, for the data movement. Accordingly, a latency between the memory device 100a and the external device, which is caused due to the data movement, may be removed or decreased.

FIG. 2 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device 100b and the memory device 100a will be mainly described, and additional description associated with components having the same reference numerals will be omitted to avoid redundancy. Also, in the case of describing other drawings, additional description associated with components having the same reference numerals will be omitted to avoid redundancy.

The processing circuit 118 may include a command decoder 118_1, an arithmetic logic block (ALU) 118_2, and a data buffer 118_3. As in the command decoder 114, the command decoder 118_1 may identify the ID of the channel CH1, may decode the command and address signals CA1, and may control the ALU 118_2 and the data buffer 118_3. The ALU 118_2 may execute various arithmetic or logic operations under control of the command decoder 118_1, such as addition, subtraction, multiplication, division, shift, AND, NAND, OR, NOR, XNOR, and XOR. The above operation may be also referred to as a “processing operation”. The data buffer 1183 may include a plurality of registers or latches. The data buffer 1183 may store data of the internal data input/output bus 119 or the common data input/output bus 102, or intermediate or final data of the arithmetic logic operation executed by the ALU 118_2. The processing circuit 128 may include a command decoder 128_1, an ALU 128_2, and the data buffer 128_3 that operate to be similar to the components 118_1 to 118_3. As in the command decoder 124, the command decoder 128_1 may identify the ID of the channel CH2, may decode the command and address signals CA2, and may control the ALU 128_2 and the data buffer 128_3. For example, a processing operation that is executed by the ALU 1182 may be identical to or different from a processing operation that is executed by the ALU 128_2.

The memory device 100b may further receive a processing command PCMD2. For example, the external device may further issue the processing command PCMD2, and the memory device 100b may receive the processing command PCMD2 after the broadcast commands BCMD1 and BCMD2. Compared to the case of the memory device 100a, the external device may further request the processing circuit 128 to perform the processing operation on moved

data by issuing the broadcast commands BCMD1 and BCMD2 and then issuing the processing command PCMD2.

The command and address buffering circuit 198 may receive the processing command PCMD2 through the channel CH2 and the pins 194 and may output the processing command PCMD2 together with the ID of the channel CH2 to the common command and address bus 101. The command decoder 128_1 may identify the ID of the channel CH2 and may decode the processing command PCMD2. The data buffer 128_3 may store the data MD transmitted from the data input/output circuit 125 through the internal data input/output bus 129 or the common data input/output bus 102. The data MD may be data that have not yet been stored in the memory cell array 123, such as data received from the core die 110, or data that are stored in the memory cell array 123 and are then read from the memory cell array 123. The ALU 128_2 may execute a processing operation with respect to the data MD of the data buffer 128_3 to generate the data PD. The data buffer 128_3 may store data PD, and the data PD may be transmitted to the data input/output circuit 125 through the internal data input/output bus 129 or the common data input/output bus 102. The data input/output circuit 125 may transmit the data PD to the memory cell array 123 under control of the command decoder 124 decoding the broadcast command BCMD2 or may transmit the data PD to the memory cell array 123 under control of the command decoder 124 decoding both the broadcast command BCMD2 and the processing command PCMD2, and the data PD may be stored in the memory cell array 123. The data PD, not the data MD, may be stored in the memory cell array 123.

In an example embodiment, the processing command PCMD2 may be merged to the broadcast command BCMD2. The broadcast command BCMD2 may include information for requesting a processing operation of the processing circuit 128 on an end location, as well as an end and the end location of the data movement. The broadcast commands BCMD1 and BCMD2 may be distinguished from each other by the order of transmission or by a different bit value(s) of a specific signal(s) of the broadcast commands BCMD1 and BCMD2.

FIG. 3 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device 100c and the memory device 100b will be mainly described. The external device may further generate external data ED. The data input/output buffering circuit 199 may receive the data input/output signals DQ2 including the external data ED through the channel CH2 and the pins 194 and may output the data input/output signals DQ2 including the external data ED to the common data input/output bus 102. The data buffer 128_3 may further receive and store the external data ED through the common data input/output bus 102. The ALU 128_2 may execute a processing operation with respect to the data MD and the external data ED to generate the data PD. The data PD may be transmitted to the memory cell array 123 by the data input/output circuit 125 under control of the command decoder 124 decoding the broadcast command BCMD2 or the processing command PCMD2. The external device may request the movement of the data MD from the core die 110 to the core die 120 and the processing operation of the data MD and the external data ED, by further inputting the external data ED to the memory device 100c. The external data ED may be based on the processing command PCMD2 or the broadcast command BCMD2.

FIG. 4 is a flowchart illustrating an operating method of memory devices of FIGS. 1 to 3. The flowchart of FIG. 4

may be applied to the memory devices **100a** to **100c**. In operation **S105**, the command and address buffering circuit **198** of the buffer die **190** may output the broadcast command **BCMD1** and the ID of the channel **CH1** received through the channel **CH1** to the common command and address bus **101**. In operation **S110**, the command decoder **114** of the core die **110** may check the ID associated with the broadcast command **BCMD1**, may identify the ID of the channel **CH1** and may decode the broadcast command **BCMD1**. In operation **S115**, the command decoder **124** of the core die **120** may check the ID associated with the broadcast command **BCMD1**, may identify the ID of the channel **CH1** and may not decode the broadcast command **BCMD1**. For example, each of the command decoders **114** and **124** may compare the previously programmed or stored ID of each of the channels **CH1** and **CH2** and the ID transmitted through the common command and address bus **101** and may decode a command depending on whether the IDs are matched. In operation **S120**, under control of the command decoder **114**, data stored in the memory cell array **113** may be read, and the data input/output circuit **115** may output the data MD to the common data input/output bus **102**.

In operation **S125**, the command and address buffering circuit **198** of the buffer die **190** may output the broadcast command **BCMD2** and the ID of the channel **CH2** received through the channel **CH2** to the common command and address bus **101**. In operation **S130**, the command decoder **114** of the core die **110** may check the ID associated with the broadcast command **BCMD2**, may identify the ID of the channel **CH2** and may not decode the broadcast command **BCMD2**. In operation **S135**, the command decoder **124** of the core die **120** may check the ID associated with the broadcast command **BCMD2**, may identify the ID of the channel **CH2** and may decode the broadcast command **BCMD2**. For example, while the data MD are output from the memory cell array **113** to the common data input/output bus **102** in operation **S120**, operation **S125** to operation **S135** may be performed. In operation **S140**, the data input/output circuit **125** of the core die **120** may receive the data MD of the common data input/output bus **102**. Depending on a next command (e.g., the processing command **PCMD2**) or the broadcast command **BCMD2**, the data input/output circuit **125** may transmit the data MD or the data PD being a result of executing the processing operation of the processing circuit **128** on the data MD.

In operation **S145**, the command and address buffering circuit **198** of the buffer die **190** may output the processing command **PCMD2** and the ID of the channel **CH2** received through the channel **CH2** to the common command and address bus **101**. In operation **S150**, the command decoder **118_1** of the core die **110** may check the ID associated with the processing command **PCMD2**, may identify the ID of the channel **CH2** and may not decode the processing command **PCMD2**. In operation **S155**, the command decoder **128_1** of the core die **120** may check the ID associated with the processing command **PCMD2**, may identify the ID of the channel **CH2** and may decode the processing command **PCMD2**. In operation **S160**, the ALU **128_2** may execute the processing operation on the data MD of the internal data input/output bus **129** or data of the common data input/output bus **102**. When the data input/output buffering circuit **199** of the buffer die **190** receives the external data ED through the channel **CH2**, the ALU **128_2** may execute the processing operation on the data MD and the external data ED of the common data input/output bus **102**. In operation **S165**, the data input/output circuit **125** may transmit the data

PD to the memory cell array **123**, and the data PD may be written to the memory cell array **123**.

In an example embodiment, an operation mode of the memory device **100a/100b/100c** may be classified as a normal mode or a processing mode (or a broadcast mode). The external device may transmit, to the memory device **100a/100b/100c**, normal commands (e.g., an active command, a precharge command, a read command, and a write command) for writing data to the memory device **100a/100b/100c** or reading data from the memory device **100a/100b/100c**, and the memory device **100a/100b/100c** may operate in the normal mode. The external device may transmit, to the memory device **100a/100b/100c**, a broadcast command or a processing command for requesting processing operations supported by the processing circuit **118/128** of the memory device **100a/100b/100c**, and the memory device **100a/100b/100c** may operate in the processing mode, not in the normal mode. The processing mode and the normal mode may be distinguished from each other depending on a kind of a command transmitted to the memory device **100a/100b/100c**.

FIG. 5 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device **100d** and the memory device **100b/100c** will be mainly described. The memory device **100b/100c** may receive the processing command **PCMD2** after the broadcast commands **BCMD1** and **BCMD2**. The command and address buffering circuit **198** of the memory device **100d** may receive the broadcast command **BCMD1** through the channel **CH1** and the pins **193**, may then receive the processing command **PCMD2** through the channel **CH2** and the pins **194**, and may then receive the broadcast command **BCMD2** through the channel **CH2** and the pins **194**. In one example embodiment, unlike the example illustrated in FIG. 5, the processing command **PCMD2** may be included in the broadcast command **BCMD2**. The command and address buffering circuit **198** may output the broadcast command **BCMD1** and the ID of the channel **CH1**, the processing command **PCMD2** and the ID of the channel **CH2**, and the broadcast command **BCMD2** and the ID of the channel **CH2** to the common command and address bus **101**. The data input/output circuit **115** may output the data MD to the common data input/output bus **102** under control of the command decoder **114** decoding the broadcast command **BCMD1**, the ALU **128_2** may execute the processing operation on the data MD of the common data input/output bus **102** under control of the command decoder **128_1** decoding the processing command **PCMD2**, the data buffer **128_3** may output the data PD to the common data input/output bus **102** or the internal data input/output bus **129**, and the data input/output circuit **125** may receive the data PD under control of the command decoder **124** decoding the broadcast command **BCMD2**. The data input/output buffering circuit **199** of the memory device **100d** may further receive the external data ED through the channel **CH2** and the pins **194**, and the ALU **128_2** may execute the processing operation on the data MD and the external data ED to generate the data PD.

FIG. 6 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device **100e** and the memory device **100d** will be mainly described. The command and address buffering circuit **198** of the memory device **100e** may receive the broadcast command **BCMD1** through the channel **CH1** and the pins **193**, may then receive a processing command **PCMD1** through the channel **CH1** and the pins **193**, and may then receive the broadcast command **BCMD2**

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through the channel CH2 and the pins 194. In one example embodiment, unlike the example illustrated in FIG. 6, the processing command PCMD1 may be included in the broadcast command BCMD1. The command and address buffering circuit 198 may output the broadcast command BCMD1 and the ID of the channel CH1, the processing command PCMD1 and the ID of the channel CH1, and the broadcast command BCMD2 and the ID of the channel CH2 to the common command and address bus 101. The data input/output circuit 115 may output the data MD to the common data input/output bus 102 under control of the command decoder 114 decoding the broadcast command BCMD1, the ALU 118_2 may execute the processing operation on the data MD of the common data input/output bus 102 under control of the command decoder 118_1 decoding the processing command PCMD1, the data buffer 118_3 may output the data PD to the common data input/output bus 102, and the data input/output circuit 125 may receive the data PD of the common data input/output bus 102 under control of the command decoder 124 decoding the broadcast command BCMD2. The data input/output buffering circuit 199 of the memory device 100e may further receive the external data ED through the channel CH1 and the pins 193, and the ALU 118_2 may execute the processing operation on the data MD and the external data ED to generate the data PD.

FIG. 7 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device 100f and the memory device 100d/100e will be mainly described.

The command and address buffering circuit 198 of the memory device 100f may receive the broadcast command BCMD1 through the channel CH1 and the pins 193, may then receive the processing command PCMD1 through the channel CH1 and the pins 193, may then receive the processing command PCMD2 through the channel CH2 and the pins 194, and may then receive the broadcast command BCMD2 through the channel CH2 and the pins 194. In one example embodiment, unlike the examples illustrated in FIG. 7, the processing command PCMD1 may be included in the broadcast command BCMD1 and the processing command PCMD2 may be included in the broadcast command BCMD2. The command and address buffering circuit 198 may output the broadcast command BCMD1 and the ID of the channel CH1, the processing command PCMD1 and the ID of the channel CH1, the processing command PCMD2 and the ID of the channel CH2, and the broadcast command BCMD2 and the ID of the channel CH2 to the common command and address bus 101. The data input/output circuit 115 may output the data MD to the common data input/output bus 102 under control of the command decoder 114 decoding the broadcast command BCMD1, the ALU 118_2 may execute the processing operation on the data MD of the common data input/output bus 102 under control of the command decoder 118_1 decoding the processing command PCMD1, the data buffer 118_3 may output the data PD to the common data input/output bus 102, the ALU 128_2 may execute the processing operation on the data PD of the common data input/output bus 102 under control of the command decoder 128_1 decoding the processing command PCMD2, the data buffer 128_3 may output the data PD to the common data input/output bus 102 or the internal data input/output bus 129, and the data input/output circuit 125 may receive the data PD under control of the command decoder 124 decoding the broadcast command BCMD2. The data input/output buffering circuit 199 of the memory device 100f may further receive the external data ED through the channel CH1 and the pins 193

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or may receive the external data ED through the channel CH2 and the pins 194; the ALU 1182 may execute the processing operation on the data MD and the external data ED to generate the data PD, or the ALU 128_2 may execute the processing operation on the data MD and the external data ED to generate the data PD.

FIG. 8 is a flowchart illustrating an operation method of a memory device of FIG. 7. The flowchart of FIG. 8 may be applied to the memory devices 100d to 100f. Operation S205 to operation S220 are substantially the same as operation S105 to operation S120. Operation S225 to operation S240 are substantially identical to operation S145 to operation S160 except that operation S225 to operation S240 relate to the channel CH1. Operation S245 to operation S260 are substantially identical to operation S145 to operation S160. Operation S265 to operation S285 are substantially identical to operation S125 to operation S140 and operation S165.

FIG. 9 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device 100g and the memory device 100a will be mainly described.

The memory device 100g may further include a core die 130. The core die 130 may include TSVs 131 and 132, a memory cell array 133, a command decoder 134, a data input/output circuit 135, a processing circuit 138, and an internal data input/output bus 139. The core die 130 may be stacked on the core die 120. As in the channels CH1 and CH2, a channel CH3 may include access (or communication) paths capable of accessing the memory device 100g and may be independent of the channels CH1 and CH2. The external device may access the memory cell array 133 or the processing circuit 138 through the channel CH3. Operations of the components 133 to 139 of the core die 130 may be similar or substantially identical to the operations of the components 113 to 119/123 to 129 of the core die 110/120 except that the core die 130 is accessible through the channel CH3 and the core die 110/120 is accessible through the channel CH1/CH2.

The buffer die 190 may further include pins 195. The pins 195 may be implemented to be substantially identical to the pins 193 except that the pins 195 are allocated to the channel CH3. As in the command and address signals CA1/CA2, the command and address buffering circuit 198 may output command and address signals CA3 to the common command and address bus 101. As in the data input/output signals DQ1/DQ2, the data input/output buffering circuit 199 may transmit data input/output signals DQ3.

The memory device 100g may further receive a broadcast command BCMD3 from the external device through the channel CH3. For example, the broadcast command BCMD3 may be transmitted to the memory device 100g together with the broadcast command BCMD2 after the broadcast command BCMD1 or may be transmitted to the memory device 100g after the broadcast command BCMD2. Alternatively, the broadcast command BCMD3 may be transmitted after the broadcast command BCMD1, and then, the broadcast command BCMD2 may be transmitted. The broadcast commands BCMD1, BCMD2, and BCMD3 may request the movement of data from the memory cell array 113 allocated to the channel CH1 to the memory cell arrays 123 and 133 respectively allocated to the channels CH2 and CH3. For example, the broadcast command BCMD3 transmitted to the memory device 100g through the channel CH3 may indicate an end and an end location of the data movement (i.e., the memory cell array 133 allocated to the channel CH3). For example, the number of memory cell arrays or channels corresponding to the end location may be

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one or more. Also, a processing command may be further merged to the broadcast command BCMD3.

After operations of the command and address buffering circuit 198, the memory cell array 113, the command decoder 114, and the data input/output circuit 115, which are performed in response to the broadcast command BCMD1, are completed, the data MD may be output to the common data input/output bus 102. After operations of the command and address buffering circuit 198, the memory cell array 123, the command decoder 124, and the data input/output circuit 125, which are performed in response to the broadcast command BCMD2, are completed, the data input/output circuit 125 may receive the data MD of the common data input/output bus 102. As in the operations of the command and address buffering circuit 198, the memory cell array 123, the command decoder 124, and the data input/output circuit 125, which are performed in response to the broadcast command BCMD2, operations of the command and address buffering circuit 198, the memory cell array 133, the command decoder 134, and the data input/output circuit 135, which are performed in response to the broadcast command BCMD3, are completed, the data input/output circuit 135 may receive the data MD of the common data input/output bus 102.

For example, the command and address buffering circuit 198 may output the command and address signals CA1 including the broadcast command BCMD1 to the common command and address bus 101, may then output the command and address signals CA2 including the broadcast command BCMD2 to the common command and address bus 101, and may then output the command and address signals CA3 including the broadcast command BCMD3 to the common command and address bus 101. The data input/output circuit 125 may receive the data MD of the common data input/output bus 102 before the data input/output circuit 135. For another example, the command and address buffering circuit 198 may output the command and address signals CA3 including the broadcast command BCMD3 to the common command and address bus 101 before the command and address signals CA2 including the broadcast command BCMD2. The data input/output circuit 135 may receive the data MD of the common data input/output bus 102 before the data input/output circuit 125.

FIG. 10 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device 100h and the memory device 100g will be mainly described.

The command and address buffering circuit 198 may output the command and address signals CA1 including the broadcast command BCMD1 to the common command and address bus 101 and may output a command and address signals, to which the broadcast command BCMD2 and the broadcast command BCMD3 are merged, to the common command and address bus 101. Logical values of the command and address signals CA2 and CA3 respectively indicating the broadcast commands BCMD2 and BCMD3 may be identical. The command and address buffering circuit 198 may merge the broadcast commands BCMD2 and BCMD3 and may output the merged broadcast command having the same logical values to the common command and address bus 101. The command and address buffering circuit 198 may merge IDs of the channels CH2 and CH3 or may set, to "Don't care", a specific bit capable of identifying the channels CH2 and CH3 from among bits of the IDs of the channels CH2 and CH3. The command and address buffering circuit 198 may output the merged ID simultaneously indicating the channels CH2 and CH3 to the common

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command and address bus 101, together with the merged broadcast command. Accordingly, the command decoders 124 and 134 may simultaneously decode the merged broadcast command, and the data input/output circuits 125 and 135 may simultaneously receive the data MD of the common data input/output bus 102.

FIG. 11 is a flowchart illustrating an operation method of a memory device of FIG. 10. The flowchart of FIG. 11 may be applied to the memory devices 100g and 100h. Operation S305, operation S310, and operation S325 may be substantially identical to operation S105, operation S110, and operation S120. Each of operation S315 and operation S320 may be substantially identical to operation S115. In operation S330, the command and address buffering circuit 198 of the buffer die 190 may merge the broadcast commands BCMD2 and BCMD3 received through the channels CH2 and CH3 and may output a merged broadcast command BCMD23 and a merged ID indicating the channels CH2 and CH3 to the common command and address bus 101. Operation S335 may be substantially identical to operation S150. Operation S340 and operation S350 may be substantially identical to operation S135 and operation S140. Operation S345 and operation S355 may be substantially identical to operation S135 and operation S140.

In an example embodiment, the memory device 100g/100h may further receive a processing command in addition to the broadcast commands BCMD1 to BCMD3. For example, as described with reference to FIGS. 2 to 4, the processing command may be transmitted to the memory device 100g/100h after the broadcast commands BCMD1 to BCMD3. For another example, as described with reference to FIGS. 5 to 8, the processing command may be transmitted to the memory device 100g/100h after the broadcast command BCMD1 is transmitted and before the broadcast commands BCMD2 and BCMD3 are transmitted. The processing command may request one of the processing circuits 118 to 138 to perform a processing operation on the data MD of the common data input/output bus 102 or the data PD being a result of a processing operation performed by another of the processing circuits 118 to 138. Two or more processing commands may be transmitted to the memory device 100g/100h, and the processing commands may request two or more of the processing circuits 118 to 138 of the core dies 110 to 130 to perform processing operations on the data MD or the data PD of the common data input/output bus 102. Also, as described with reference to FIGS. 5 to 8, the external data ED may be further transmitted to the memory device 100g/100h together with the processing command. The processing command may request one of the processing circuits 118 to 138 of the core dies 110 to 130 to perform a processing operation on the data MD, the data PD, or the external data ED of the common data input/output bus 102. Example embodiments are not limited to those described with reference to FIGS. 1 to 11. For example, the number of core dies stacked on the buffer die 190 may be one or more. For example, a start location of the data movement may be another memory cell array allocated to another channel without limitation to the memory cell array 113 allocated to the channel CH1, and an end location of the data movement may be one or more memory cell arrays. For example, one or more processing commands may be transmitted to the buffer die 190 together with broadcast commands or may be included in broadcast commands, one or more processing operations may be performed on the data MD/PD of the common data input/output bus 102 or the external data ED.

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FIG. 12 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device 200a and the memory device 100b will be mainly described. The memory device 200a may include core dies 210 and 220 and a buffer die 290. Each of the core dies 110 to 130 of the memory devices 100a to 100h described above may be allocated to one channel, but each of the core dies 210 and 220 may be allocated to two channels. Of course, each of the core dies 210 and 220 may be allocated to two or more channels.

The core die 210 may include TSVs 211_1 and 212_1, a memory cell array 213_1, a command decoder 214_1, a data input/output circuit 215_1, a processing circuit 218_1 including a command decoder 218_11, an ALU 218_12, and a data buffer 218_13, and an internal data input/output bus 219_1. The components 211_1 to 219_1 may be allocated to the channel CH1 so as to be accessed through the channel CH1 and may be implemented to be identical to the components 111 to 119 of the memory devices 100a to 100h. The core die 210 may further include TSVs 211_2 and 212_2, a memory cell array 213_2, a command decoder 214_2, a data input/output circuit 215_2, a processing circuit 218_2 including a command decoder 218_21, an ALU 218_22, and a data buffer 218_23, and an internal data input/output bus 219_2. The components 211_2 to 219_2 may be allocated to the channel CH2 so as to be accessed through the channel CH2 and may be implemented to be identical to the components 111 to 119 of the memory devices 100a to 100h.

The core die 220 may include TSVs 221_3 and 222_3, a memory cell array 223_3, a command decoder 224_3, a data input/output circuit 225_3, a processing circuit 228_3 including a command decoder 228_31, an ALU 228_32, and a data buffer 228_33, and an internal data input/output bus 229_3. The components 221_3 to 229_3 may be allocated to the channel CH3 so as to be accessed through the channel CH3 and may be implemented to be identical to the components 111 to 119 of the memory devices 100a to 100h. The core die 220 may further include TSVs 221_4 and 222_4, a memory cell array 223_4, a command decoder 224_4, a data input/output circuit 225_4, a processing circuit 228_4 including a command decoder 228_41, an ALU 228_42, and a data buffer 228_43, and an internal data input/output bus 229_4. The components 221_4 to 229_4 may be allocated to the channel CH4 so as to be accessed through the channel CH4 and may be implemented to be identical to the components 111 to 119 of the memory devices 100a to 100h.

The buffer die 290 may include TSVs 291_1 and 292_1, pins 293_1 and 294_1, a command and address buffering circuit 298_1 and a data input/output buffering circuit 299_1. The components 291_1 to 294_1, 298_1, and 299_1 may be allocated to the channels CH1 and CH3 and may be implemented to be identical to the components 191 to 199 of the memory devices 100a to 100h. The buffer die 290 may further include TSVs 291_2 and 292_2, pins 293_2 and 294_2, a command and address buffering circuit 298_2 and a data input/output buffering circuit 299_2. The components 291_2 to 294_2, 298_2 and 299_2 may be allocated to the channels CH2 and CH4 and may be implemented to be identical to the components 191 to 199 of the memory devices 100a to 100h. For example, unlike the example illustrated in FIG. 12, the command and address buffering circuits 298_1 and 298_2 may be merged to one circuit, and the data input/output buffering circuits 299_1 and 299_2 may also be merged to one circuit.

The memory device 200a may include a common command and address bus 201_1 being common to the channels CH1 and CH3, a common command and address bus 201_2

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being common to the channels CH2 and CH4, a common data input/output bus 202_1 being common to the channels CH1 and CH3, and a common data input/output bus 202_2 being common to the channels CH2 and CH4. The external device may transmit broadcast commands for requesting the data movement between the memory cell arrays 213_1 and 223_3 allocated to the channels CH1 and CH3 to the memory device 200a and may transmit broadcast commands for requesting the data movement between the memory cell arrays 213_2 and 223_4 allocated to the channels CH2 and CH4 to the memory device 200a. The external device may transmit a processing command(s) for requesting a processing operation(s) on the data MD, the data PD, or the external data ED by the movement between the memory cell arrays 213_1 and 223_3 allocated to the channels CH1 and CH3 to the memory device 200a through at least one of the channels CH1 and CH3 and may transmit a processing command(s) for requesting a processing operation(s) on the data MD, the data PD, or the external data ED by the movement between the memory cell arrays 213_2 and 223_4 allocated to the channels CH2 and CH4 to the memory device 200a through at least one of the channels CH2 and CH4. As in the memory devices 100g and 100h, the memory device 200a may further include another core die that is stacked on the core die 220, is implemented to be identical to the core dies 210 and 220, and is allocated to channels CH5 and CH6. In an example embodiment, processing operations that are executed by the ALUs 218_12 and 228_32 may be identical to or different from each other. Processing operations that are executed by the ALUs 218_22 and 228_42 may be identical to or different from each other. Processing operations that are executed by the ALUs 218_12 and 218_22 may be identical to or different from each other. Processing operations that are executed by the ALUs 228_32 and 228_42 may be identical to or different from each other.

FIG. 13 illustrates a block diagram of a memory device according to another example embodiment. A difference between a memory device 200b and the memory device 200a will be mainly described. The core die 210 may further include a common data input/output bus 212_12 that electrically connects the common data input/output buses 202_1 and 202_2 and is common to the channels CH1 and CH2. For example, the internal data input/output bus 212_12 may be electrically connected to the TSVs 212_1 and 212_2 and with the internal data input/output buses 219_1 and 219_2. The core die 220 may further include a common data input/output bus 222_34 that electrically connects the common data input/output buses 202_1 and 202_2 and is common to the channels CH3 and CH4. For example, the internal data input/output bus 222_34 may be electrically connected to the TSVs 222_3 and 222_4 and with the internal data input/output buses 229_3 and 229_4. Compared with the memory device 200a, the memory device 200b may further include the common data input/output bus 212_12 and 222_34, and thus, the memory device 200b may further support the data movement between the channels CH1 and CH2, between the channels CH1 and CH4, between the channels CH3 and CH4, and between the channels CH3 and CH2, in addition to the data movement between the channels CH1 and CH3 and between the channels CH2 and CH4. The external device may transmit broadcast commands for requesting the data movement between the memory cell arrays 213_1, 213_2, 223_3, and 223_4 allocated to the channels CH1 to CH4 to the memory device 200b through the channels CH1 to CH4. The external device may transmit a processing command(s) for requesting a processing operation(s) on the data MD, the data PD,

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or the external data ED by the movement between the memory cell arrays **213_1**, **213_2**, **223_3**, and **223_4** allocated to the channels CH1 to CH4 to the memory device **200b** through at least one of the channels CH1 to CH4. Also, as in the memory devices **100g** and **100h**, the memory device **200b** may further include another core die that is stacked on the core die **220**, is implemented to be identical to the core dies **210** and **220**, and is allocated to the channels CH5 and CH6.

FIG. 14 illustrates a block diagram of a memory area of a core die included in a memory device according to an example embodiment. A memory area **300** may indicate the remaining area other than processing areas where the processing circuits **118** to **138**, **218_1**, **218_2**, **228_3**, and **228_4** of the core dies **110** to **130** and **210** to **220** are disposed and TSV areas where the TSVs **111**, **112**, **121**, **122**, **211_1**, **212_1**, **211_2**, **212_2**, **221_3**, **222_3**, **221_4**, and **222_4** of the core dies **110** to **130** and **210** to **220** are disposed. The memory area **300** may include a bank **303**, a row decoder **301**, a column decoder **302**, a command decoder **304**, an address demultiplexer **308**, a write driver **309**, an input/output sense amplifier **310**, and a data input/output circuit **305**. The bank **303** may be a unit for dividing each of the above memory cell arrays **113**, **123**, **133**, **213_1**, **213_2**, **223_3**, and **223_4**. The number of banks **303** may be one or more, and a plurality of banks **303** may be allocated to a channel.

The command decoder **304** may correspond to each of the above command decoders **114**, **124**, **134**, **214_1**, **214_2**, **224_3**, and **224_4**, may decode a command (e.g., a broadcast command, an active command, a precharge command, a read command, a write command, and a refresh command) included in command and address signals CA, and may control components of the memory area **300**. The row decoder **301** may select a word line(s) WL of the bank **303** corresponding to a row address provided from the address demultiplexer **308**. The column decoder **302** may select a column selection line(s) of the bank **303** corresponding to a column address provided from the address demultiplexer **308** and bit lines connected to the column selection line(s). The address demultiplexer **308** may provide the row address to the row decoder **301** and the column address to the column decoder **302**. The write driver **309** may write the write data of the data input/output circuit **305** to memory cells selected by the row decoder **301** and the column decoder **302**. The input/output sense amplifier **310** may read data from the selected memory cells and may provide the read data to the data input/output circuit **305**.

The data input/output circuit **305** may correspond to each of the above data input/output circuits **115**, **125**, **135**, **215_1**, **215_2**, **225_3**, and **225_4**. The data input/output circuit **305** may include a write circuit **306** and a read circuit **307**. The write circuit **306** may receive data included in the data input/output signals DQ of the common data input/output bus **102/202_1/202_2** based on a write data strobe signal WDQS, may parallelize the received data, and may provide the parallelized data to the write driver **309**. The read circuit **307** may serialize data from the input/output sense amplifier **310** and may output the data input/output signals DQ including the serialized data to the common data input/output bus **102/202_1/202_2** based on a read data strobe signal RDQS. Instead of a bidirectional data strobe signal DQS, the write data strobe signal WDQS and the read data strobe signal RDQS may be used to capture the data input/output signals DQ. For another example, instead of the write data strobe signal WDQS and the read data strobe signal RDQS, the bidirectional data strobe signal DQS may be used

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to capture the data input/output signals DQ. In any case, the (write/read) data strobe signals WDQS/RDQS/DQS may be bidirectionally transmitted between the data input/output circuit **115/125/135/215_1/215_2/225_3/225_4** and the external device through the common data input/output bus **102/202_1/202_2**, the data input/output buffering circuit **199/299_1/299_2**, the pins **193/194/195/293_1/294_1/293_2/294_2**, and the channels CH1/CH2/CH3/CH4.

FIG. 15 illustrates an electronic device according to an example embodiment. An electronic device **1000** (or a computing/electronic system) may include a memory device **1100**, an interposer **1200**, and a system on chip **1300**. The memory device **1100** may include core dies **1110** to **1180** and a buffer die **1190**. Each of the core dies **1110** to **1180** may correspond to each of the above core dies **110** to **130**, **210**, and **220** and may be identically manufactured, and the buffer die **1190** may correspond to each of the above buffer dies **190** and **290**. The number of core dies **1110** to **1180** is not limited to the example illustrated in FIG. 15. Each of the core dies **1110** to **1180** may include a memory area **1183** corresponding to the memory area **300** and a processing area **1188** where the processing circuit **118/128/138/218_1/218_2/228_3/228_4** is disposed. The buffer die **1190** may include a physical layer **1194** (hereinafter referred to as a "PHY"), and the PHY **1194** may include the components **193** to **199**, **293_1** to **299_1**, and **293_2** to **299_2** described above. The core dies **1110** to **1180** and the buffer die **1190** may be electrically interconnected through a plurality of TSVs **1101** and a plurality of micro bumps **1102**. The TSVs **1101** may include the above TSVs **111**, **112**, **121**, **122**, **131**, **132**, **211_1**, **212_1**, **211_2**, **212_2**, **221_3**, **222_3**, **221_4**, and **222_4**.

In an example embodiment, the memory device **1100** may be a general-purpose DRAM device, such as a double data rate dynamic random access memory (DDR SDRAM), a mobile DRAM device, such as a low power double data rate (LPDDR) SDRAM, a graphics DRAM device, such as a graphics double data rate (GDDR) synchronous graphics dynamic random access memory (SGDRAM), or a DRAM device, which provides a high capacity and a high bandwidth, such that Wide I/O, a high bandwidth memory (HBM), HBM2, HBM3, or a hybrid memory cube (HMC). The interposer **1200** may connect the memory device **1100** and the system on chip **1300**. The interposer **1200** may provide physical paths which connect the PHY **1194** of the memory device **1100** and a PHY **1340** of the system on chip **1300** and are formed of conductive materials for an electrical connection. The system on chip **1300** may correspond to the external device described above. The system on chip **1300** may execute applications, which the electronic device **1000** supports, by using the memory device **1100** and may be also referred to as an "application processor (AP)". The system on chip **1300** may include the PHY **1340** that is electrically connected to the PHY **1194** of the buffer die **1190** through the interposer **1200**. The system on chip **1300** may store data to the memory device **1100** or may read data from the memory device **1100**. The system on chip **1300** may generate various commands (e.g., commands associated with a read or write operation of the memory device **1100**, the broadcast commands BCMD1 to BCMD3, and the processing commands PCMD1 and PCMD2) described with reference to FIGS. 1 to 14 and may transmit the generated commands to the memory device **1100**.

FIG. 16 is a diagram illustrating an electronic device according to another example embodiment. An electronic device **2000** may include a memory device **2100** including core dies **2110** to **2180** and a buffer die **2190**, and a system

on chip **2300** including a PHY **2340**, and each of the core dies **2110** to **2180** may include a memory area **2183** and a processing area **2188** electrically interconnected through TSVs **2201** and micro bumps **2202** and respectively corresponding to the memory area **1183** and the processing area **1188**. The memory device **2100** and the system on chip **2300** may correspond to the memory device **1100** and the system on chip **1300**, respectively. The memory device **2100** may be disposed on the system on chip **2300**, and the system on chip **2300** may further include the TSVs **2201** that are used to implement electrical connections with the memory device **2100**.

FIG. **17** illustrates a block diagram of system on chips of FIGS. **15** and **16**. A system on chip **3300** may correspond to the system on chips **1300** and **2300** described above, and may include a processor **3310**, a cache memory **3320**, a memory controller **3330**, and a PHY **3340**. A bus **3350** may provide a communication path between the processor **3310**, the cache memory **3320**, the memory controller **3330**, and the PHY **3340**. The processor **3310** may execute various software (e.g., an application program, an operating system, a file system, and a device driver) loaded to the cache memory **3320**. The processor **3310** may include a homogeneous multi-core or a heterogeneous multi-core. For example, the processor **3310** may include at least one or more of a central processing unit (CPU), an image signal processing unit (ISP), a digital signal processing unit (DSP), a graphics processing unit (GPU), a vision processing unit (VPU), and a neural processing unit (NPU), and the number of processors **3310** may be one or more. An application program, an operating system, a file system, a device driver, etc. for driving the electronic device **1000/2000** may be loaded to the cache memory **3320**. For example, the cache memory **3320** may be an SRAM device that has a faster data input/output speed than the memory device **1100/2100**. The memory controller **3330** may access the memory device **1100/2100** in a direct memory access (DMA) manner. The memory controller **3330** may include a command queue **3331**, a command scheduler **3332**, a read data queue **3333**, and a write data queue **3334**. The command queue **3331** may store commands and addresses that are generated by the processor **3310** or are generated under control of the processor **3310**. A command and an address stored in the command queue **3331** may be provided to the PHY **3340** under control of the command scheduler **3332**. The command scheduler **3332** may adjust an order of commands and addresses stored in the command queue **3331**, a time point when a command(s) and an address(es) are input to the command queue **3331**, a time point when a command(s) and an address(es) are output from the command queue **3331**, etc. The read data queue **3333** may store read data that the memory device **1100/2100** transmits through the PHY **3340** in response to the read command. The read data stored in the read data queue **3333** may be provided to the cache memory **3320** and may be processed by the processor **3310**. The write data queue **3334** may store write data to be stored to the memory device **1100/2100**. The write data stored to the write data queue **3334** by the write command may be transmitted to the memory device **1100/2100** through the PHY **3340**. The components **3331** to **3334** may be implemented in the system on chip **3300** in the form of hardware, software, or a combination thereof. The PHY **3340** may include a clock generator **3341**, a command and address generator **3342**, a data receiver **3343**, and a data transmitter **3344**. The clock generator **3341** may generate a clock CK to be output to the memory device **1100/2100**, and the number of clocks CK may correspond to the number of channels between the

system on chip **3300** and the memory device **1100/2100**. The command and address generator **3342** may receive a command or an address from the command queue **3331** and may transmit a command CMD or an address ADD to the memory device **1100/2100**. For example, the command CMD may be one of various commands (e.g., commands associated with a read or write operation of the memory device **1100**, the broadcast commands BCMD1 to BCMD3, and the processing commands PCMD1 and PCMD2) described with reference to FIGS. **1** to **14**. The data receiver **3343** may receive read data of the data input/output signal DQ based on the read data strobe signal RDQS (or DQS) from the memory device **1100/2100**. The data receiver **3343** may provide the received read data to the read data queue **3333**. The data transmitter **3344** may receive write data from the write data queue **3334**. The data transmitter **3344** may transmit the received write data to the memory device **1100/2100** based on the write data strobe signal WDQS (or DQS).

A memory device according to an example embodiment may execute the data movement between core dies based on a broadcast command from an external device, and thus, a latency between the memory device and the external device due to the data movement may be decreased.

While example embodiments have been described, it will be apparent to those of ordinary skill in the art that various changes and modifications may be made thereto without departing from the spirit and scope of the inventive concept as set forth in the following claims.

What is claimed is:

1. A memory device comprising:

a buffer die configured to receive a first command from an external device through a first channel, and receive a second command and external data from the external device through a second channel; and

a plurality of core dies stacked on the buffer die and each of the plurality of core dies connected to the buffer die through a plurality of through-silicon vias (TSVs), the plurality of core dies includes a first core die and a second core die,

wherein the first core die comprises a first processing circuit, a first memory cell array, a first command decoder circuit configured to decode the first command transmitted through a first TSV among the plurality of TSVs, and a first data input/output circuit configured to output first data stored in the first memory cell array to the second core die through a second TSV among the plurality of TSVs based on the first command,

wherein the second core die comprises a second processing circuit, a second memory cell array, a second command decoder circuit configured to decode the second command transmitted through the first TSV, and a second input/output circuit configured to receive the first data of the first memory cell array through the second TSV,

wherein the second processing circuit is configured to receive the external data and the first data and generate second data by performing a processing operation on the first data and the external data based on the second command, and

wherein the buffer die comprises a first pin configured to receive the first command and first external data through the first channel, and a second pin configured to receive the second command and second external data through the second channel.

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2. The memory device of claim 1, wherein the first external data and the second external data are transmitted to at least one of the plurality of core dies through the second TSV.

3. The memory device of claim 1, wherein the first command comprises a first broadcast command and a first processing command, and the second command comprises a second broadcast command and a second processing command.

4. The memory device of claim 3, wherein the second processing circuit is configured to receive the second processing command through the first TSV and the second external data from the buffer die through the second TSV and receive the first data from the first memory cell array through the second TSV, and

wherein the second processing circuit is configured to perform a processing operation on the second external data and the first data based on the second processing command.

5. The memory device of claim 3, wherein the first broadcast command is configured to indicate a start of data movement, and the second broadcast command is configured to indicate an end of the data movement.

6. The memory device of claim 5, wherein the first broadcast command is transmitted to the first core die prior to the second broadcast command being transmitted to the second core die.

7. The memory device of claim 5, wherein a bit of the first broadcast command indicates the start of the data movement, and a bit of the second broadcast command indicates the end of the data movement.

8. The memory device of claim 3, wherein the first processing circuit comprises a first processing command decoder, a first arithmetic logic block (ALU), and a first data buffer, and

wherein the second processing circuit comprises a second processing command decoder, a second ALU, and a second data buffer.

9. The memory device of claim 8, wherein the first processing circuit is configured to perform a first processing operation, and the second processing circuit is configured to perform a second processing operation, wherein the first processing operation and the second processing operation are different.

10. The memory device of claim 9, wherein the second processing command decoder is configured to decode the second processing command received from the buffer die through the first TSV,

wherein the second ALU is configured to perform the second processing operation on the second external data based on the second processing command to generate the second data, and

wherein the second data buffer is configured to receive the second data from the second ALU and output the second data.

11. The memory device of claim 10, wherein the second data buffer is configured to output the second data to the second input/output circuit of the second core die to be stored in the second memory cell array based on the second broadcast command.

12. The memory device of claim 10, wherein the second data buffer is configured to output the second data to the buffer die through the second TSV.

13. The memory device of claim 9, wherein each of the first processing operation and the second processing operation

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comprises at least one from among addition, subtraction, multiplication, division, shift, AND, NAND, OR, NOR, XNOR, and XOR.

14. A memory device comprising:

a buffer die configured to receive a first broadcast command and a processing command from an external device through a first channel, receive a second broadcast command from the external device through a second channel, receive external data from the external device through the first channel and the second channel, transmit the first broadcast command, the second broadcast command and the processing command through a first through silicon via (TSV) and transmit the external data through a second TSV; and

a plurality of core dies stacked on the buffer die, the plurality of core dies including a first core die allocated to the first channel and a second core die allocated to the second channel,

wherein the first core die comprises a first memory cell array, a first command decoder configured to decode the first broadcast command transmitted through the first TSV, and a first data input/output circuit configured to output first data of the first memory cell array through the second TSV,

wherein the second core die comprises a processing circuit, a second memory cell array, a second command decoder configured to decode the second broadcast command transmitted through the first TSV, and a second data input/output circuit configured to receive the first data of the first memory cell array from the first core die through the second TSV,

wherein the processing circuit of the second core die is configured to receive the processing command through the first TSV, receive the first data and the external data through the second TSV, and generate second data by performing a processing operation on the first data and the external data based on the processing command, and

wherein the buffer die comprises a first pin configured to receive a first command and first external data through the first channel, and a second pin configured to receive a second command and second external data through the second channel.

15. The memory device of claim 14, wherein the first external data and the second external data are transmitted to at least one of the plurality of core dies through the second TSV.

16. The memory device of claim 14, wherein the first command comprises the first broadcast command and a first processing command, and the second command comprises the second broadcast command and a second processing command.

17. The memory device of claim 16, wherein the processing circuit is configured to receive the second processing command through the first TSV and the second external data from the buffer die through the second TSV and receive the first data from the first memory cell array through the second TSV, and

wherein the processing circuit is configured to perform a processing operation on the second external data and the first data based on the second processing command.

18. The memory device of claim 16, wherein the first broadcast command is configured to indicate a start of data movement, and the second broadcast command is configured to indicate an end of the data movement.

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19. The memory device of claim **18**, wherein the first broadcast command is transmitted to the first core die prior to the second broadcast command being transmitted to the second core die.

20. The memory device of claim **18**, wherein a bit of the first broadcast command indicates the start of the data movement, and a bit of the second broadcast command indicates the end of the data movement.

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